



US 20250271461A1

(19) **United States**

(12) **Patent Application Publication**
Hirakawa et al.

(10) **Pub. No.: US 2025/0271461 A1**

(43) **Pub. Date: Aug. 28, 2025**

(54) **CAPACITIVE KNOB SENSING SYSTEM AND METHOD TO DETECT INITIAL STATES USING A BASELINE FRAME**

(71) Applicant: **Synaptics Incorporated**, San Jose, CA (US)

(72) Inventors: **Tomohiro Hirakawa**, Chiba (JP); **Shao-Hua Lee**, Taipei (TW); **Kuang-Lun Lin**, Taipei (TW); **Chia Yu Liu**, New Taipei City (TW); **Hsiao-Lan Hsu**, Taipei (TW); **Yu-Hung Li**, Taipei (TW)

(21) Appl. No.: **19/043,843**

(22) Filed: **Feb. 3, 2025**

Related U.S. Application Data

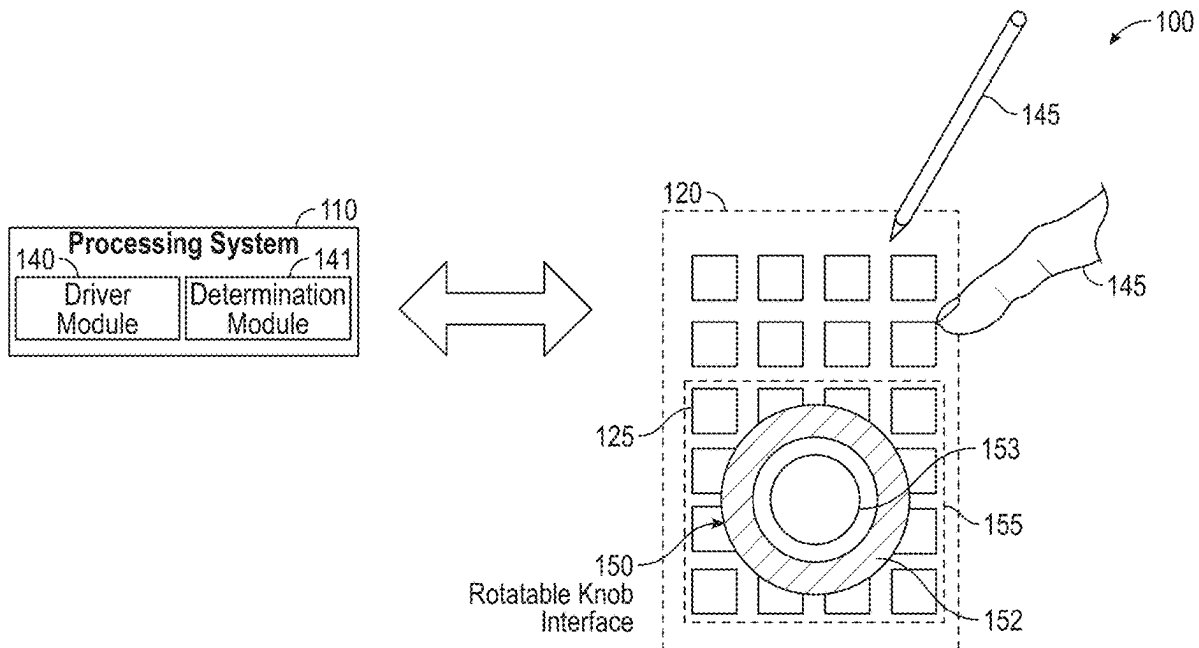
(60) Provisional application No. 63/557,766, filed on Feb. 26, 2024.

Publication Classification

(51) **Int. Cl.**
G01P 13/04 (2006.01)
G06F 3/0362 (2013.01)
G06F 3/038 (2013.01)
(52) **U.S. Cl.**
CPC **G01P 13/04** (2013.01); **G06F 3/0362** (2013.01); **G06F 3/038** (2013.01)

(57) **ABSTRACT**

A method for determining a rotational direction of a knob interface is provided. The method comprises: obtaining a baseline analog-to-digital (ADC) frame, a first ADC frame, and a second ADC frame; determining an initial state of the rotatable knob interface using the baseline ADC frame, the first ADC frame, and the second ADC frame; based on the rotatable knob interface being rotated from the initial state to a first rotational state, obtaining first rotational state resulting signals indicating the first rotational state of the rotatable knob interface; determining the rotational direction of the rotatable knob interface using the determined initial state of the rotatable knob interface and the first rotational state resulting signals; and adjusting a setting using the determined rotational direction.



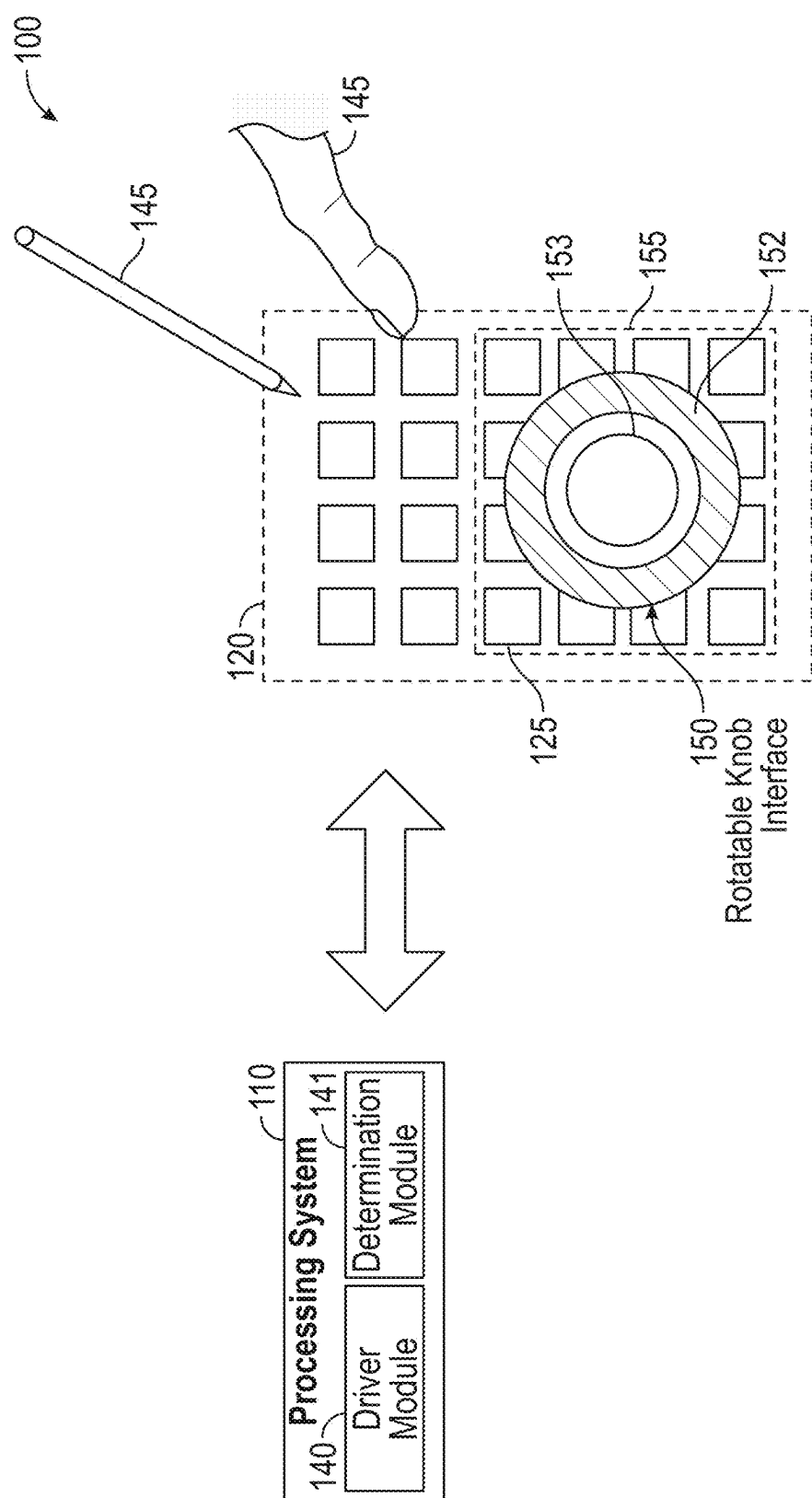


Figure 1

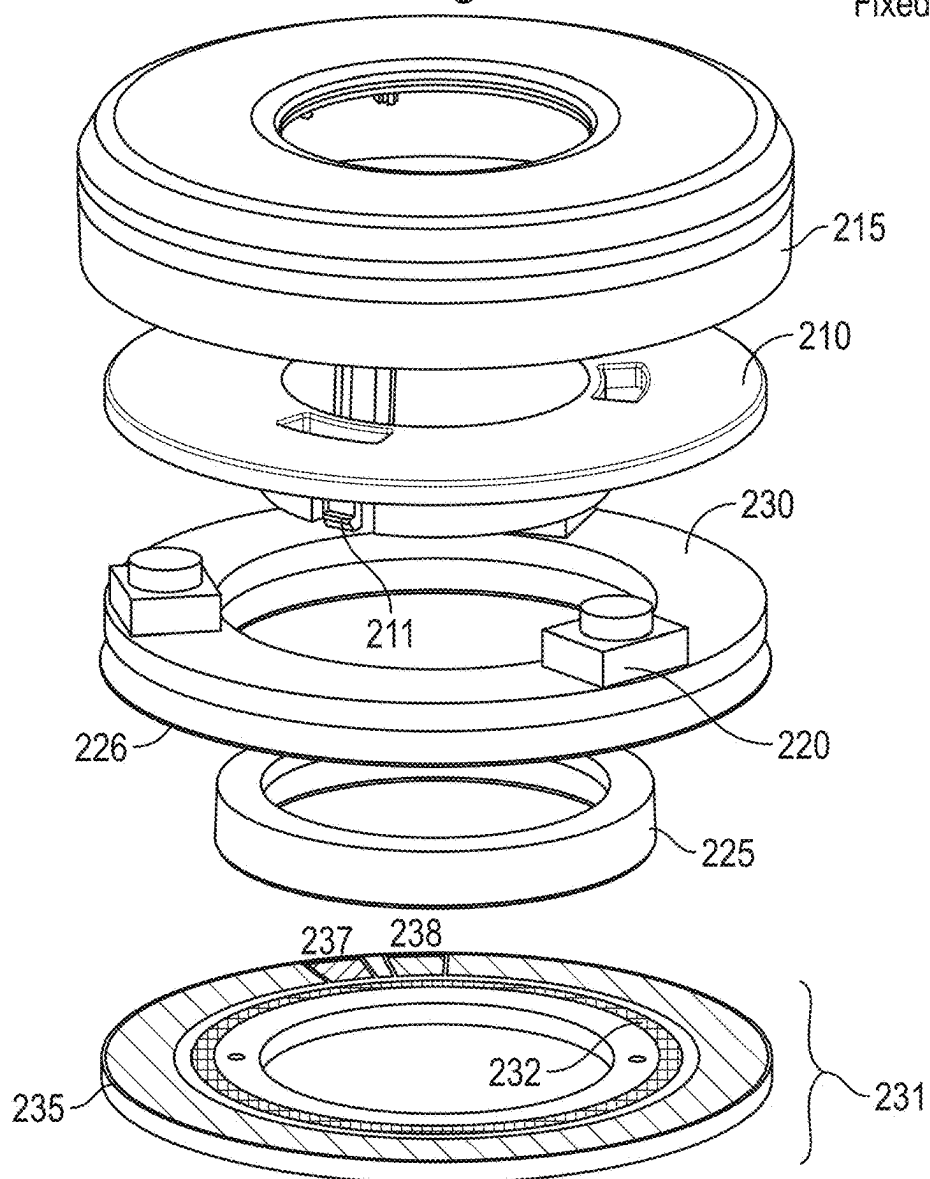
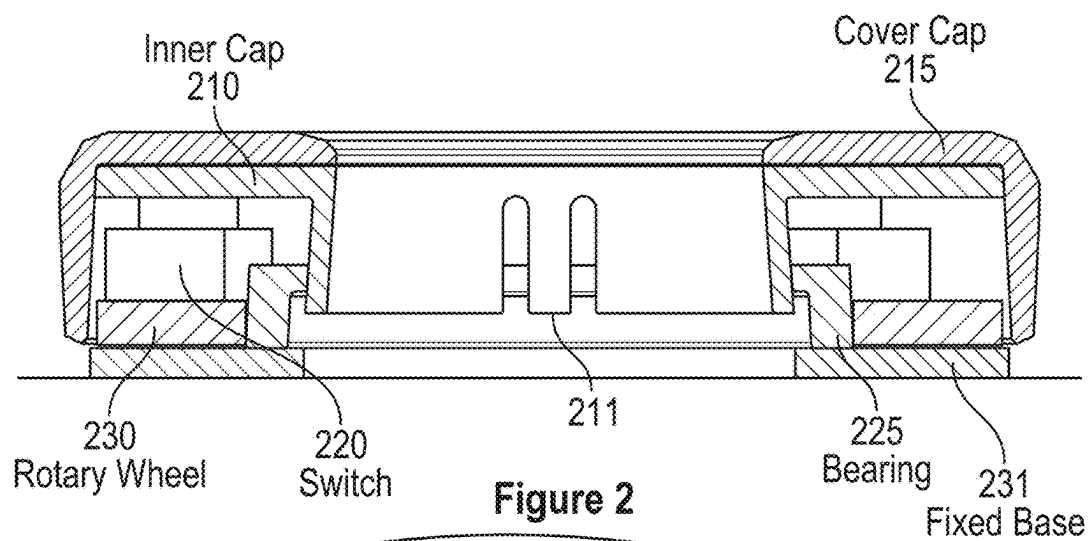


Figure 3

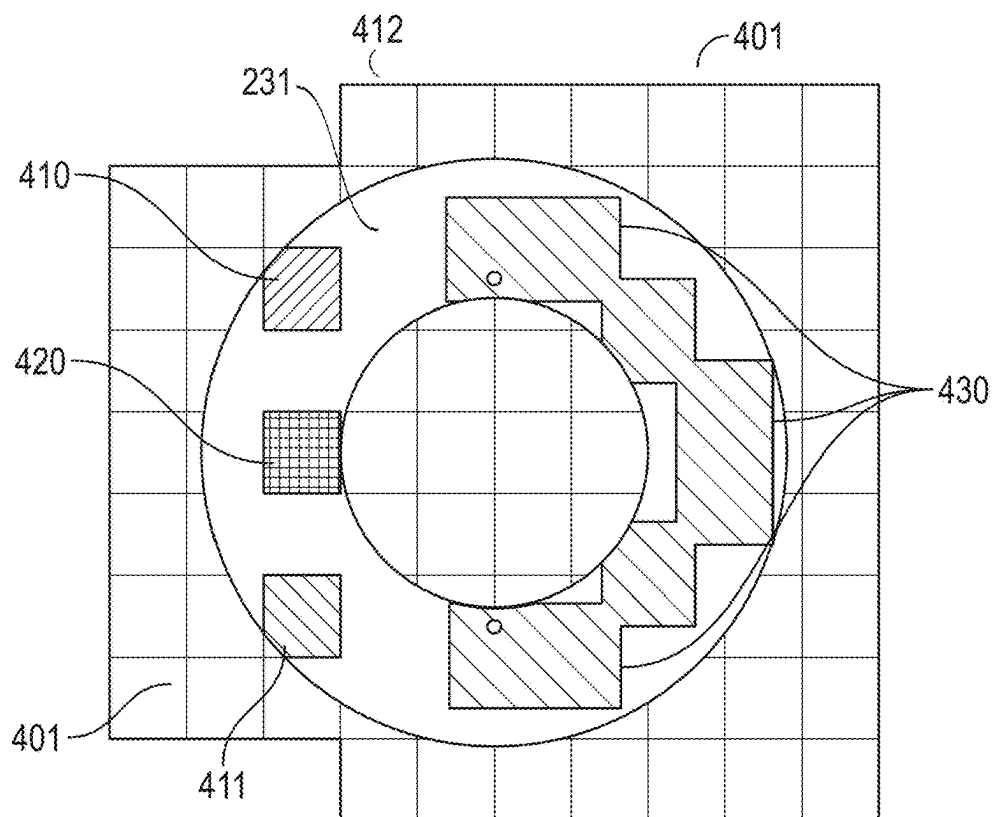


Figure 4A

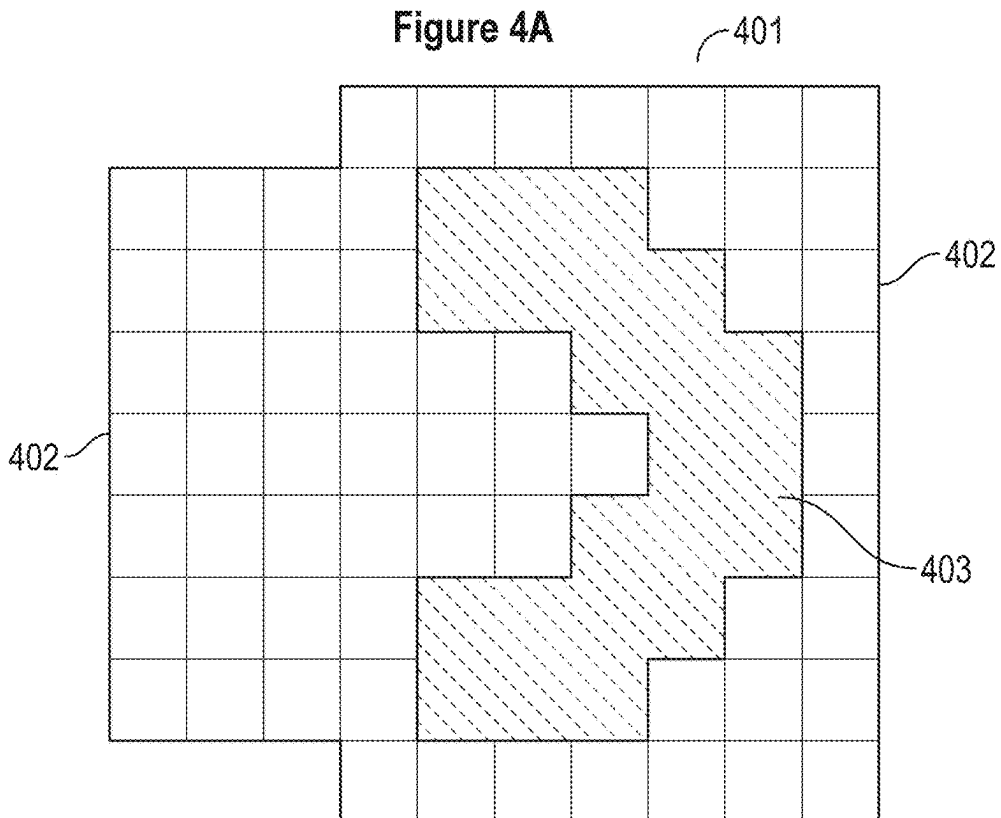


Figure 4B

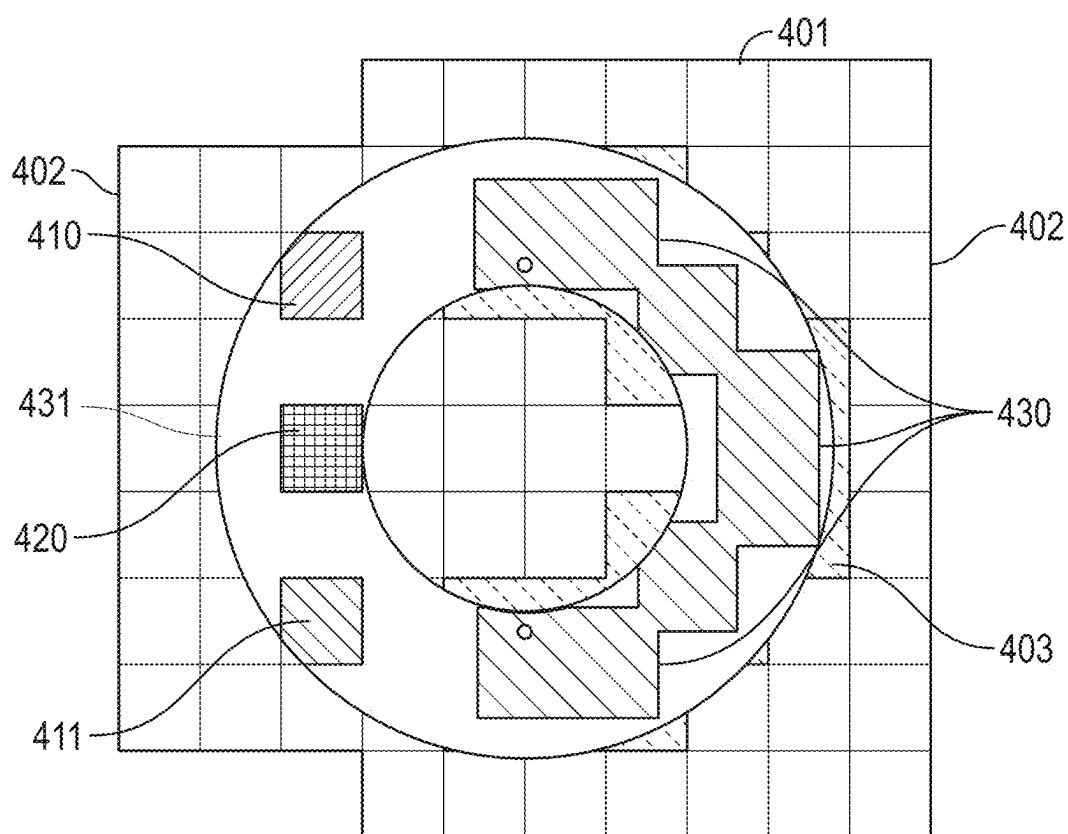


Figure 4C

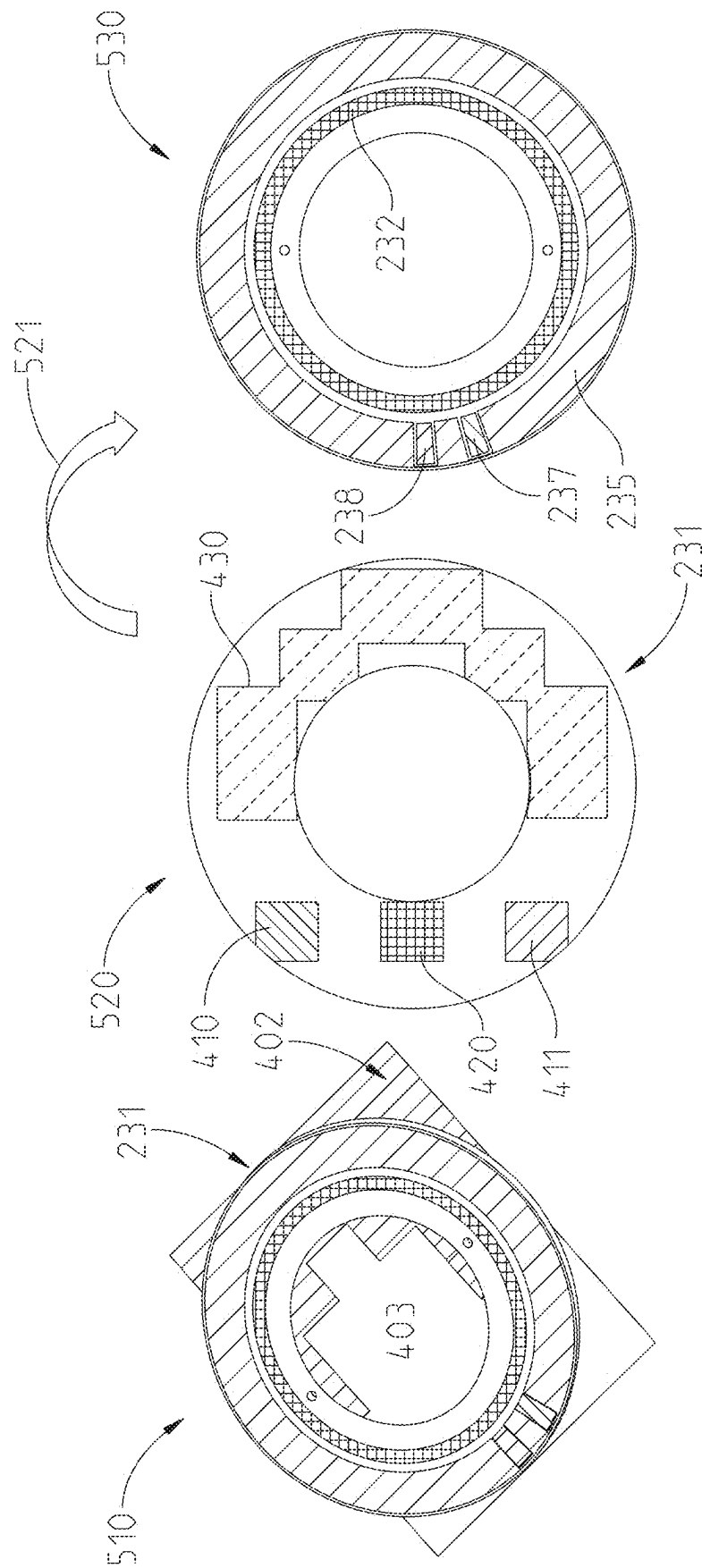


Figure 5

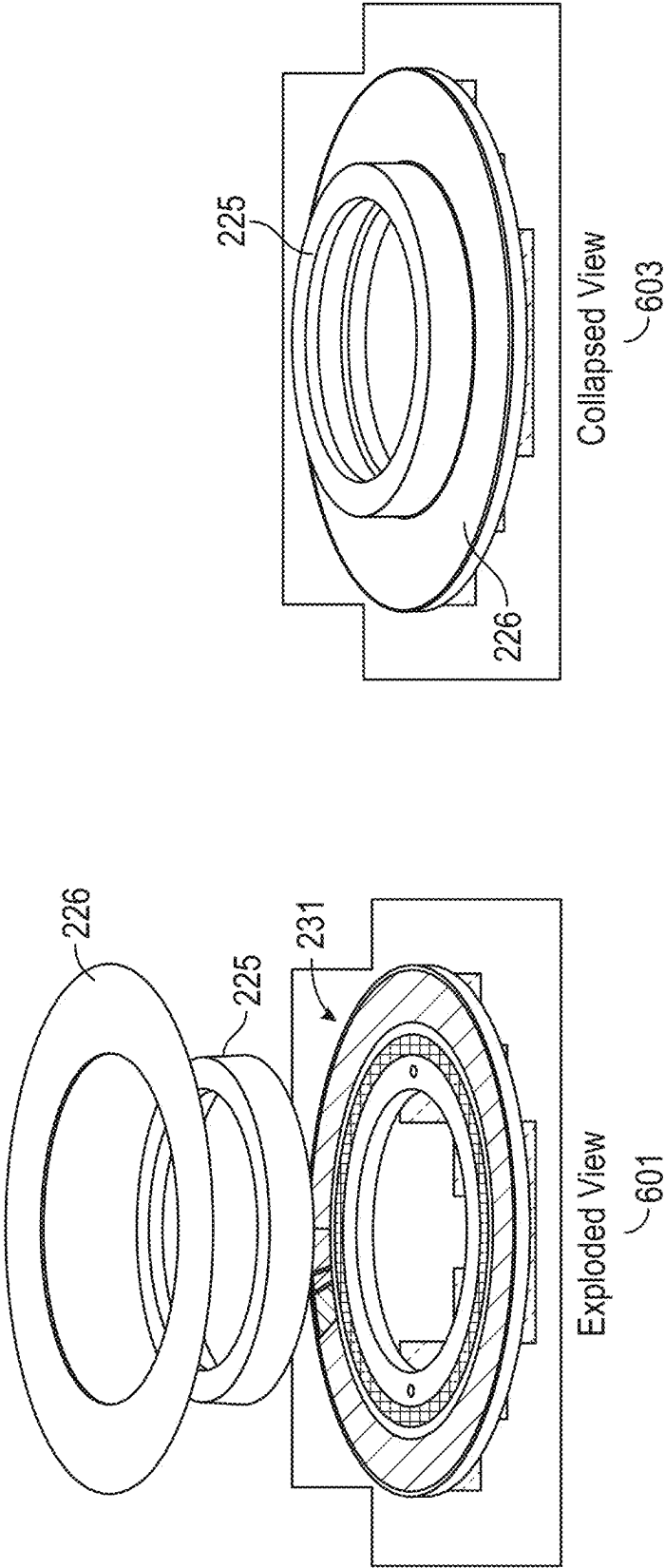


Figure 6A

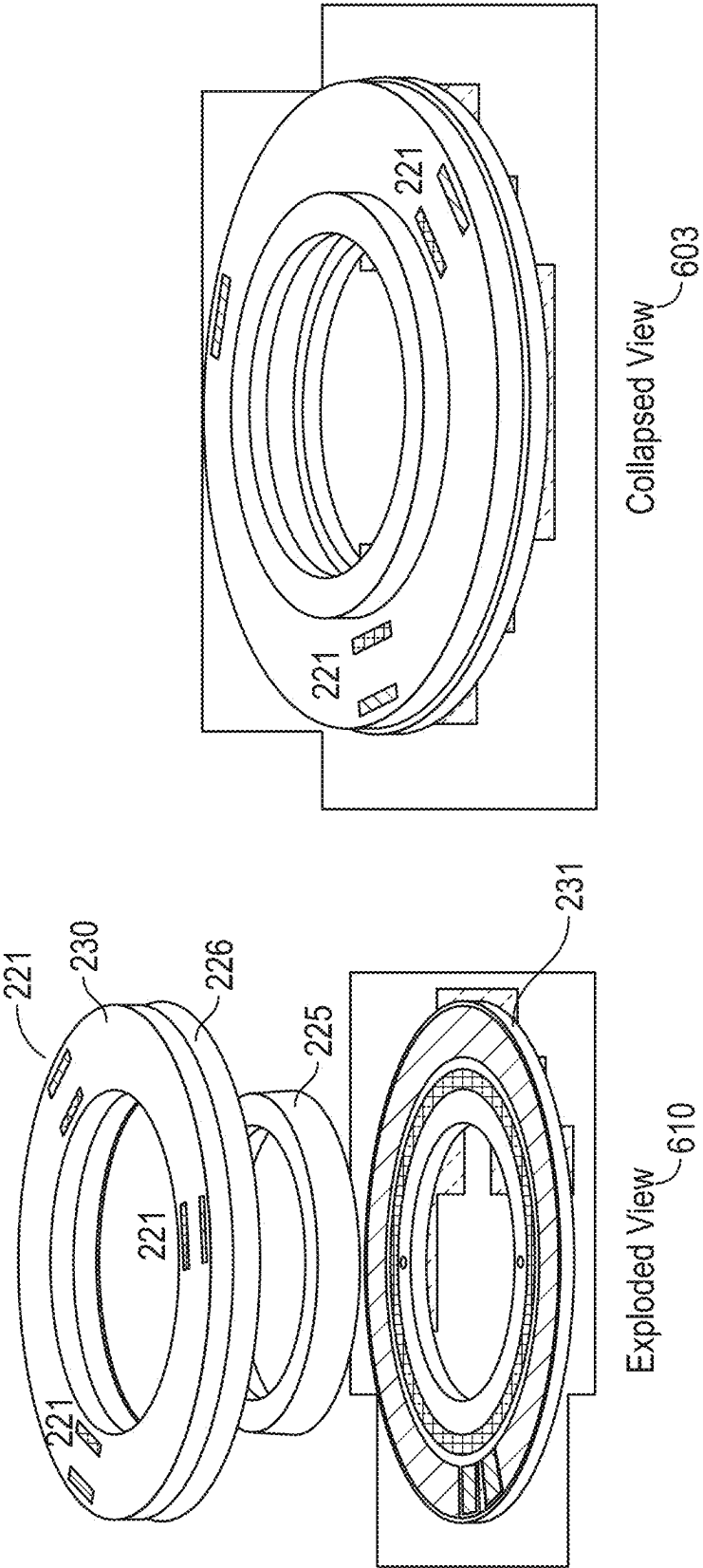


Figure 6B

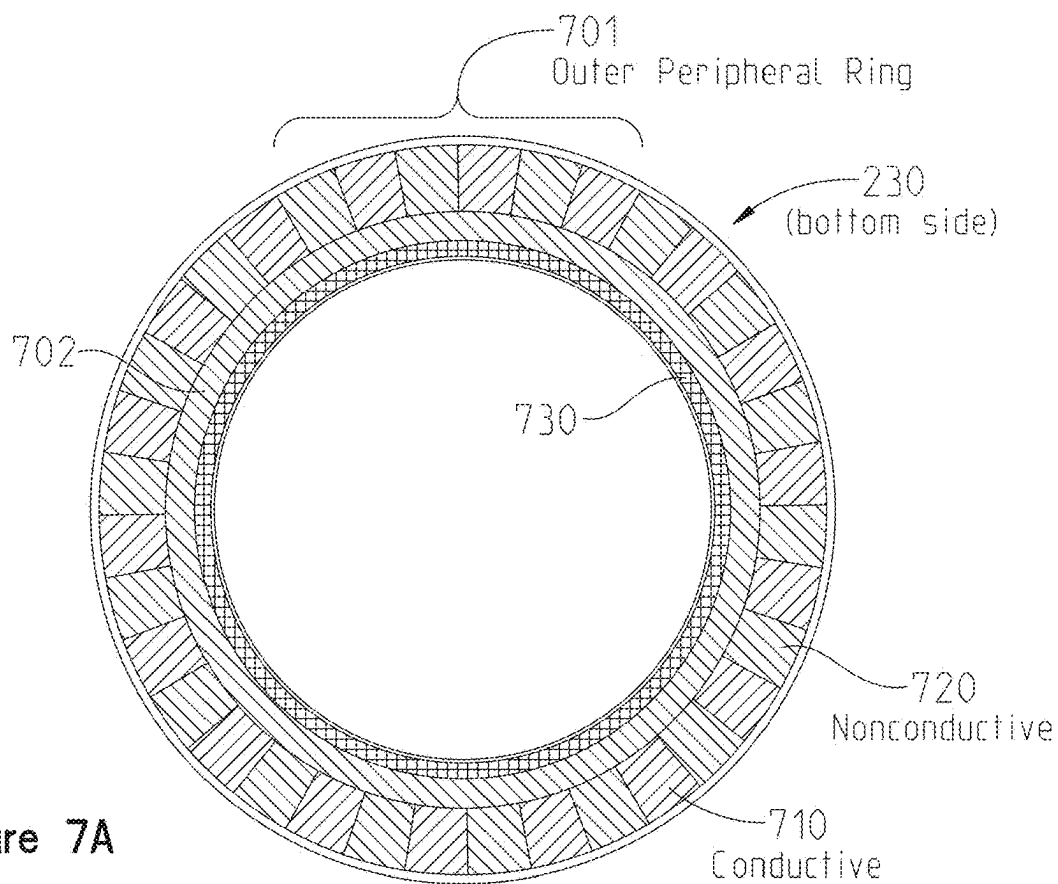


Figure 7A

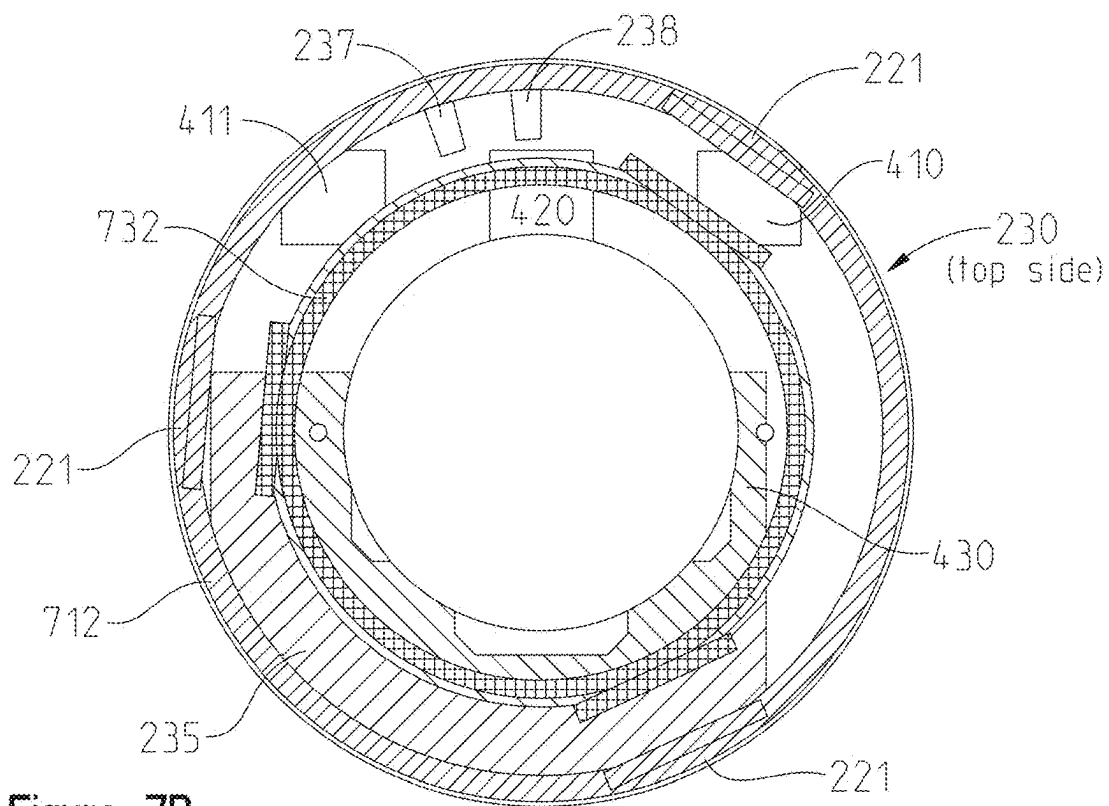


Figure 7B

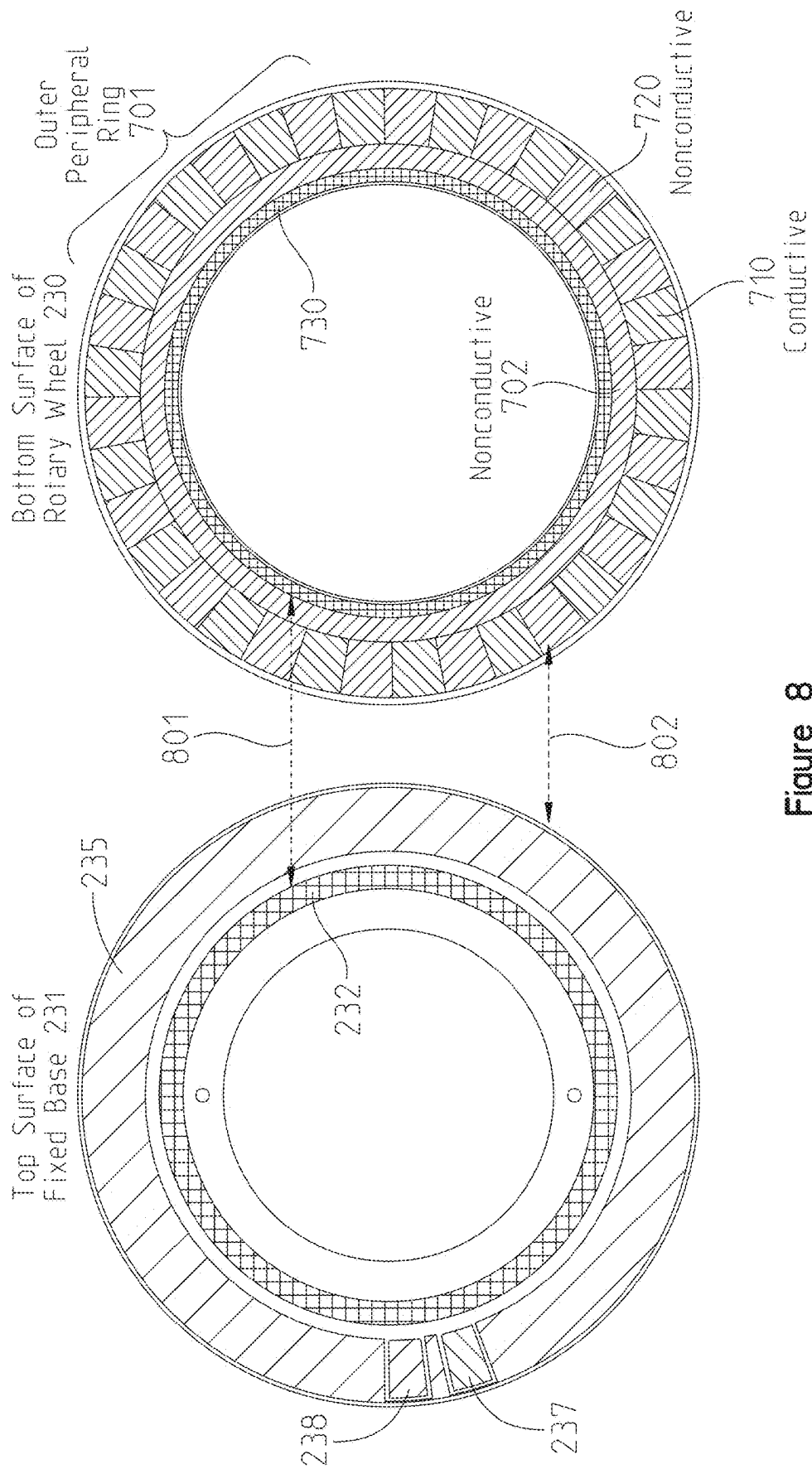


Figure 8

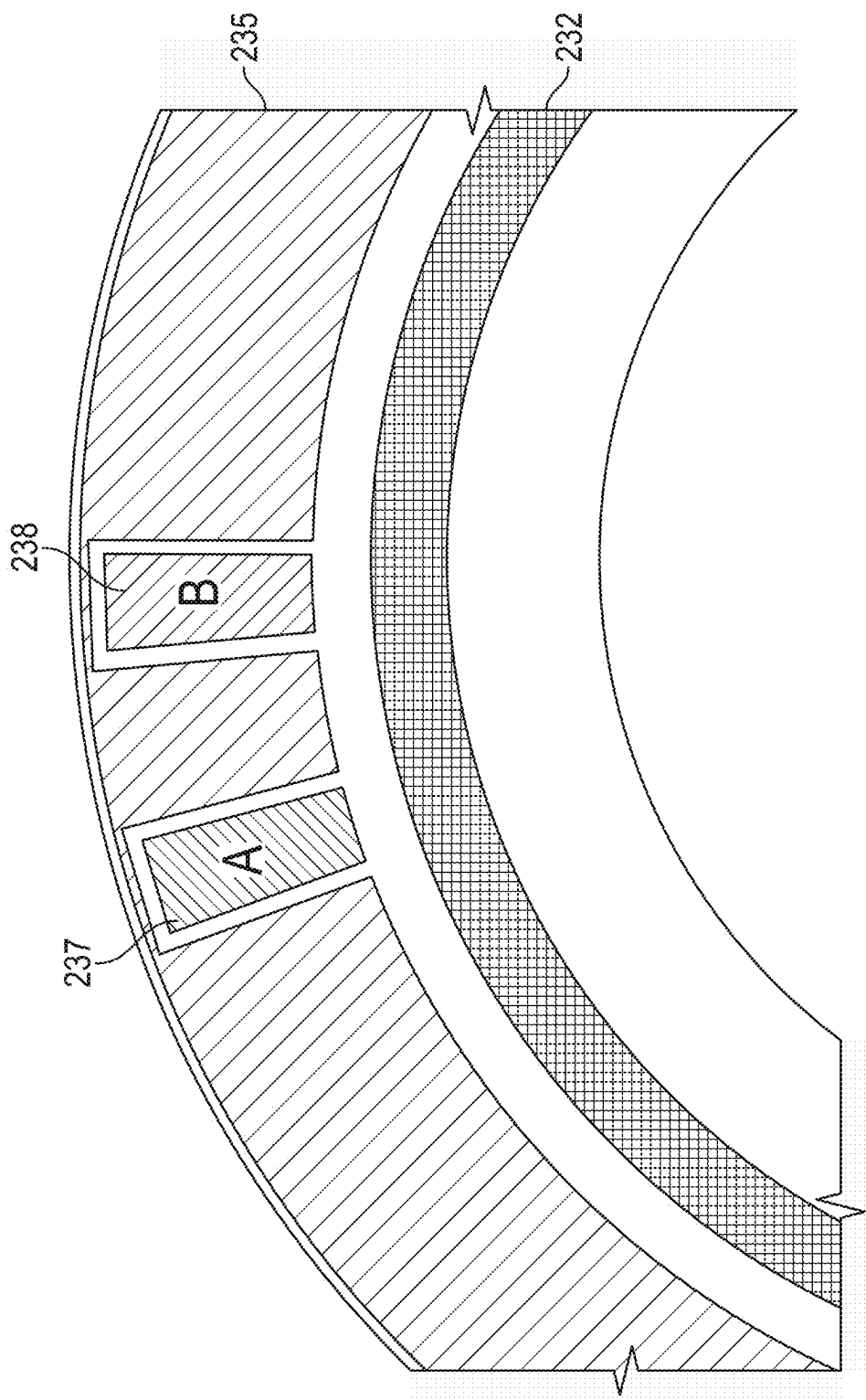
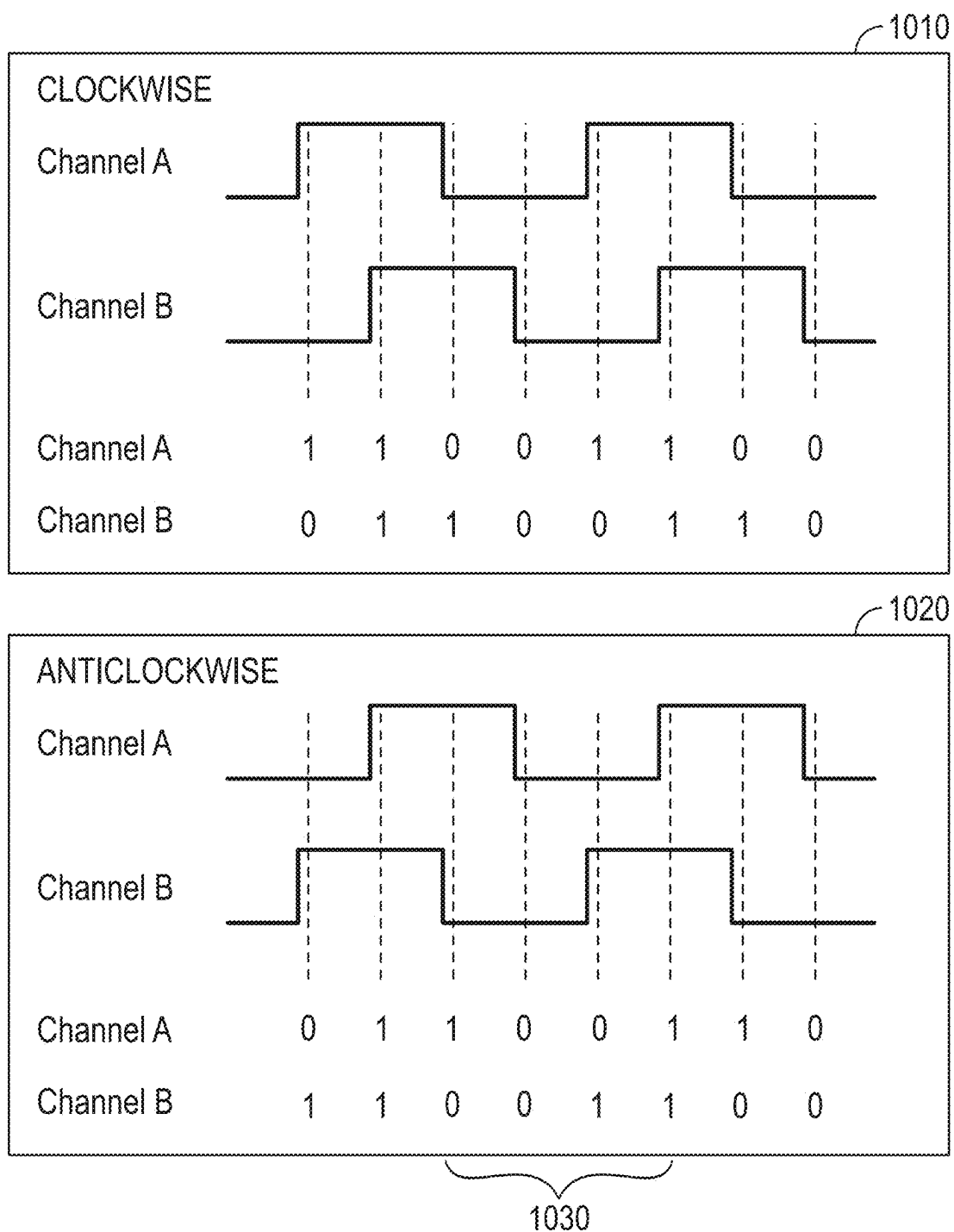


Figure 9



Sequence Shown in Figure 11A

Figure 10

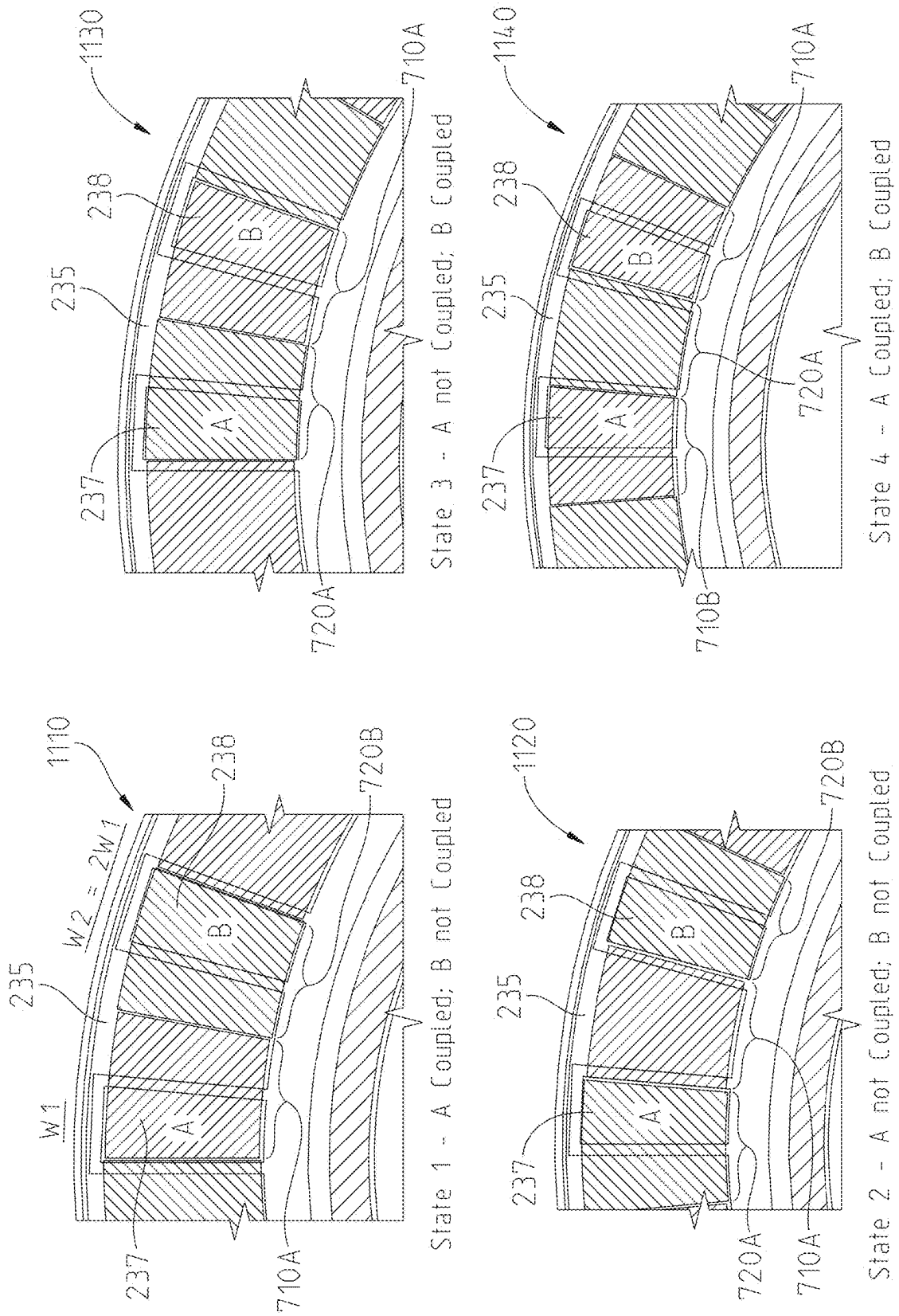


Figure 11A

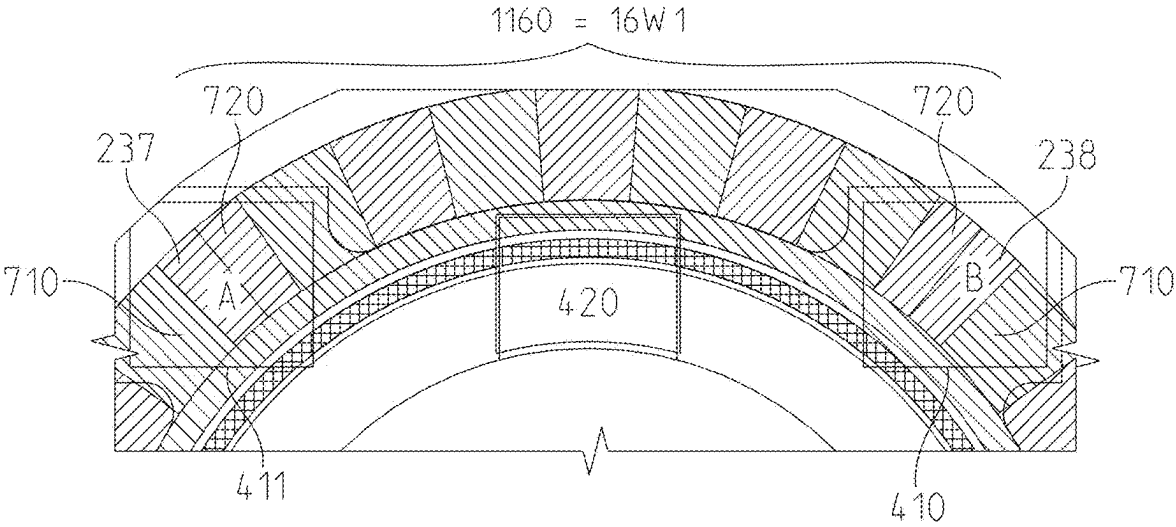
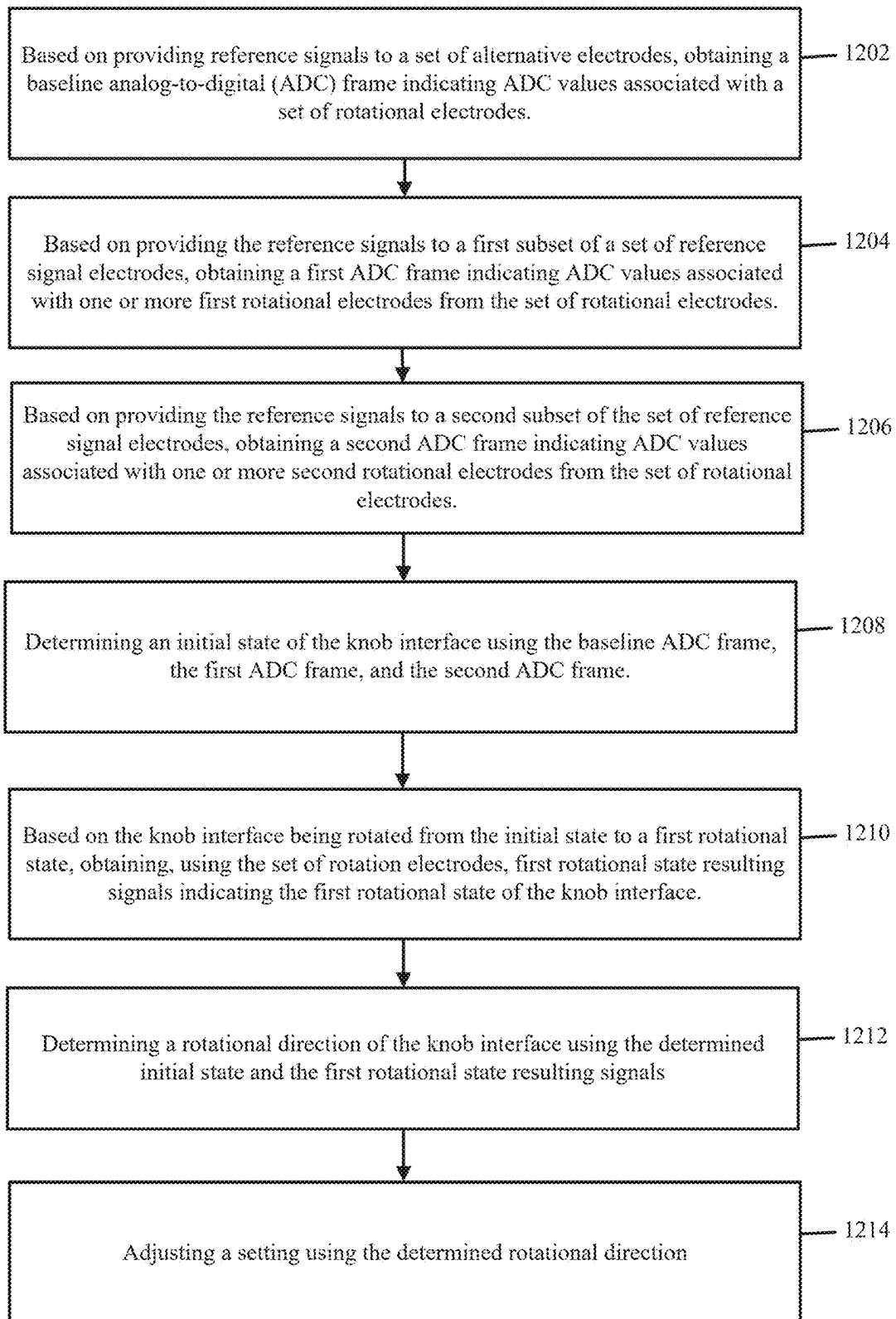
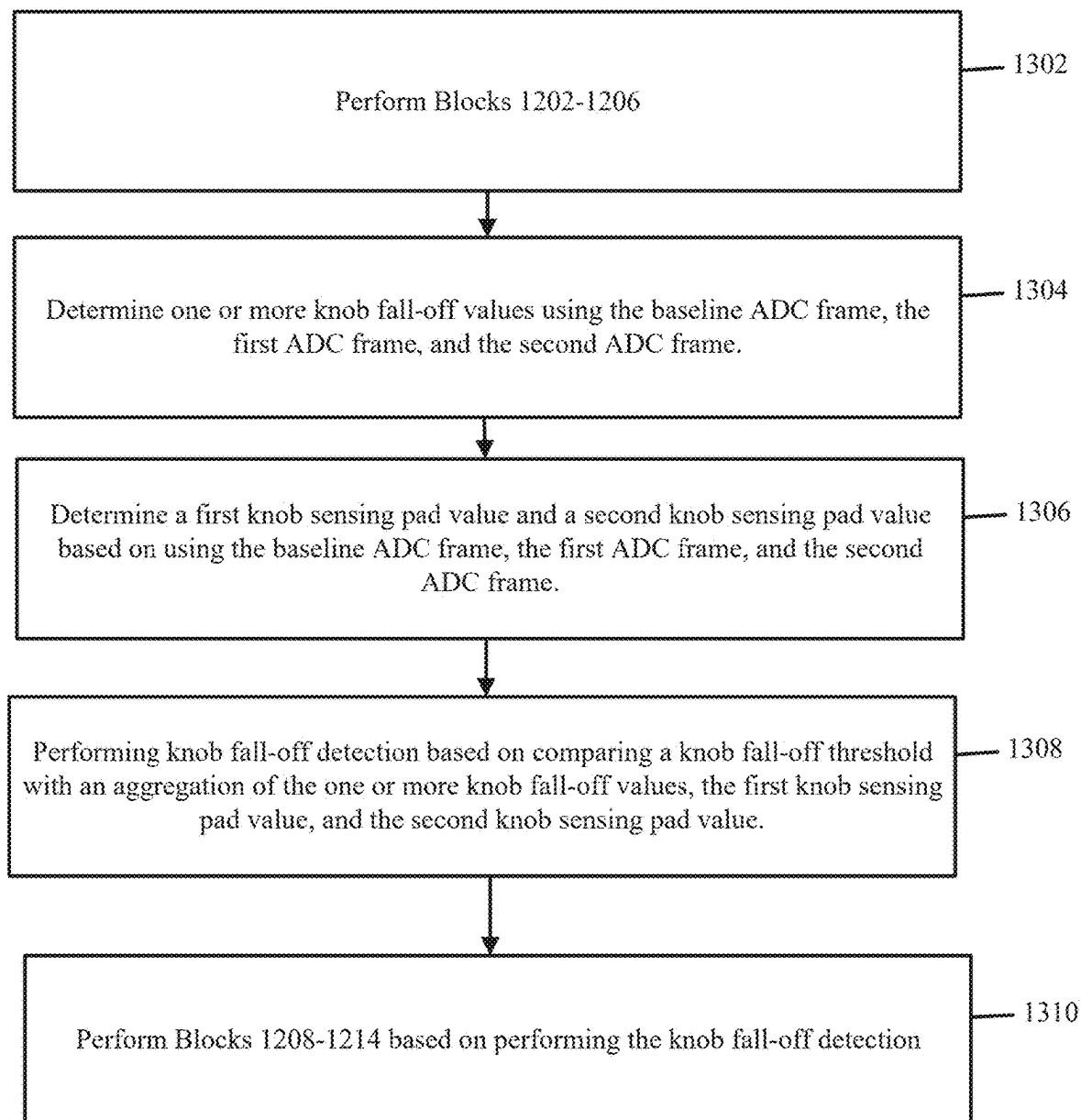


Figure 11B



1200

Figure 12



1300
Figure 13

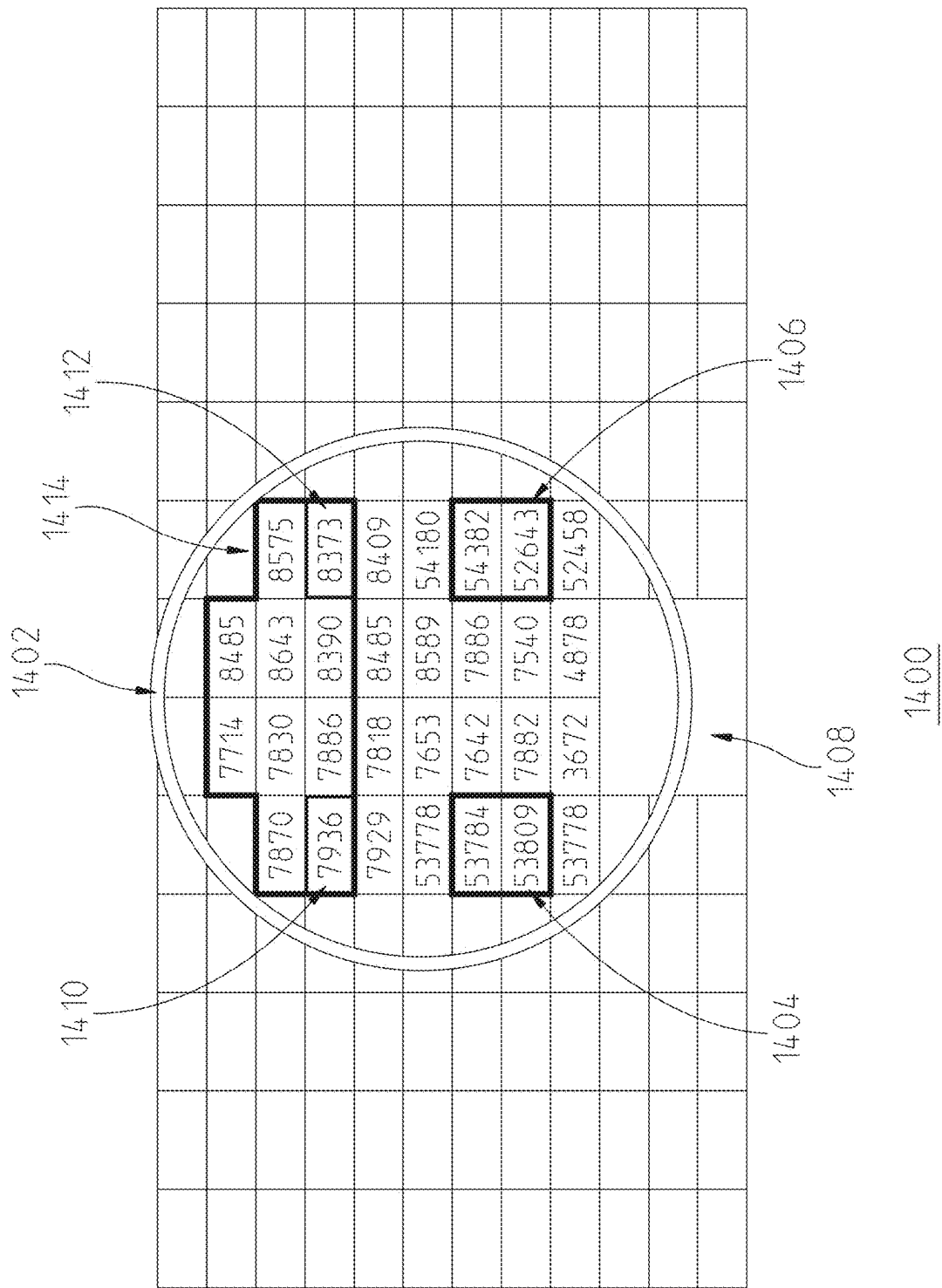


Figure 14

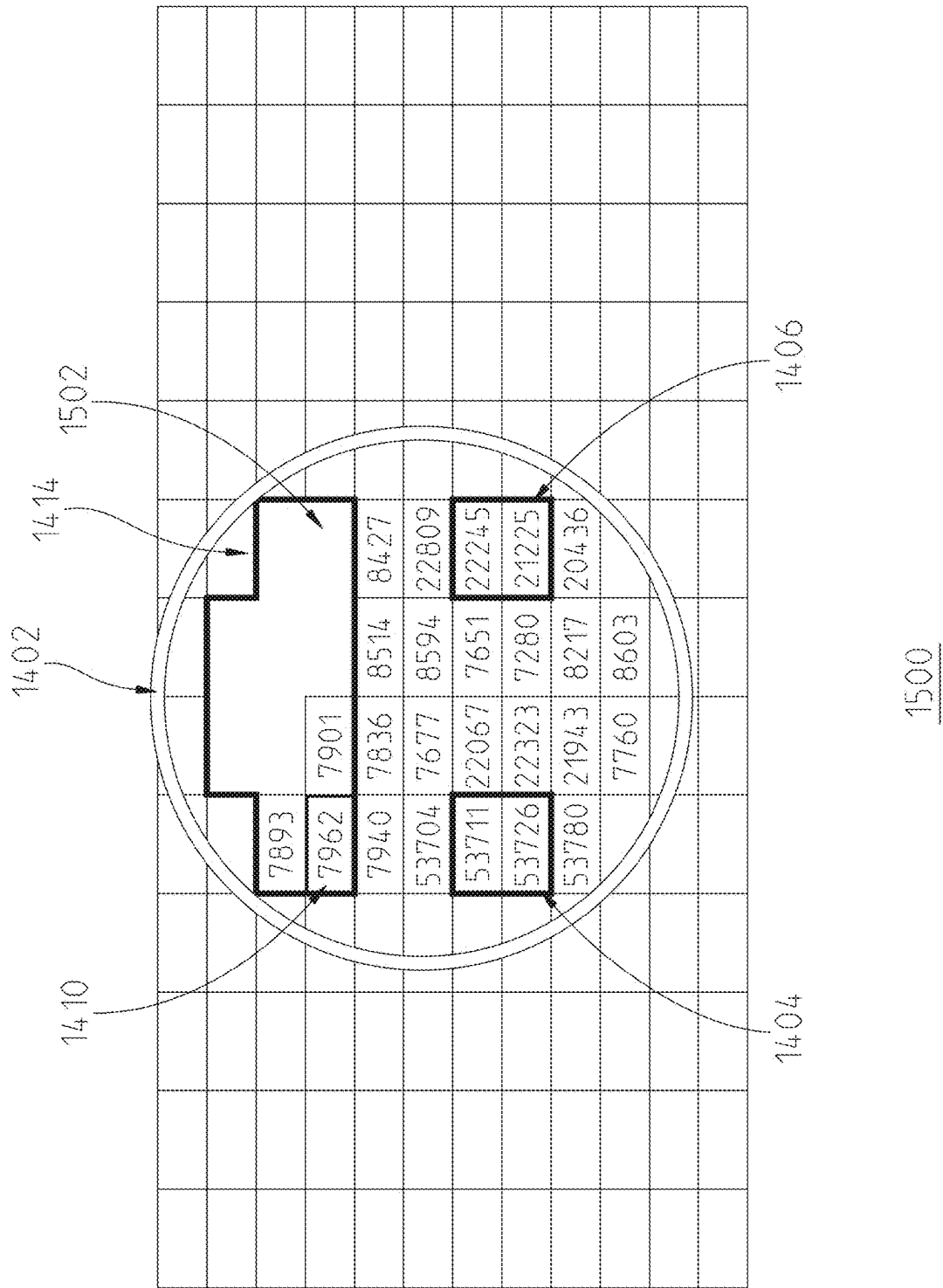


Figure 15

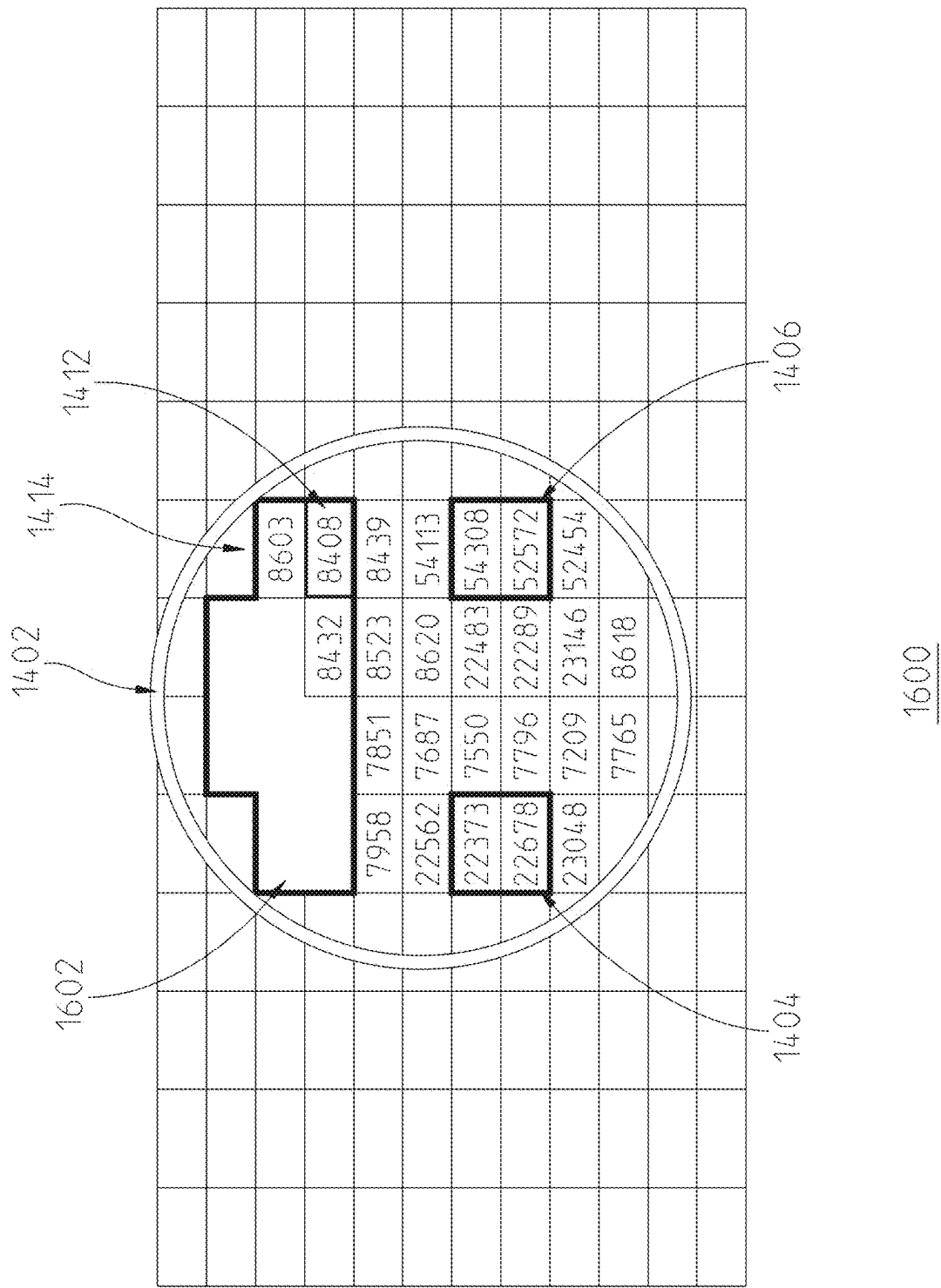
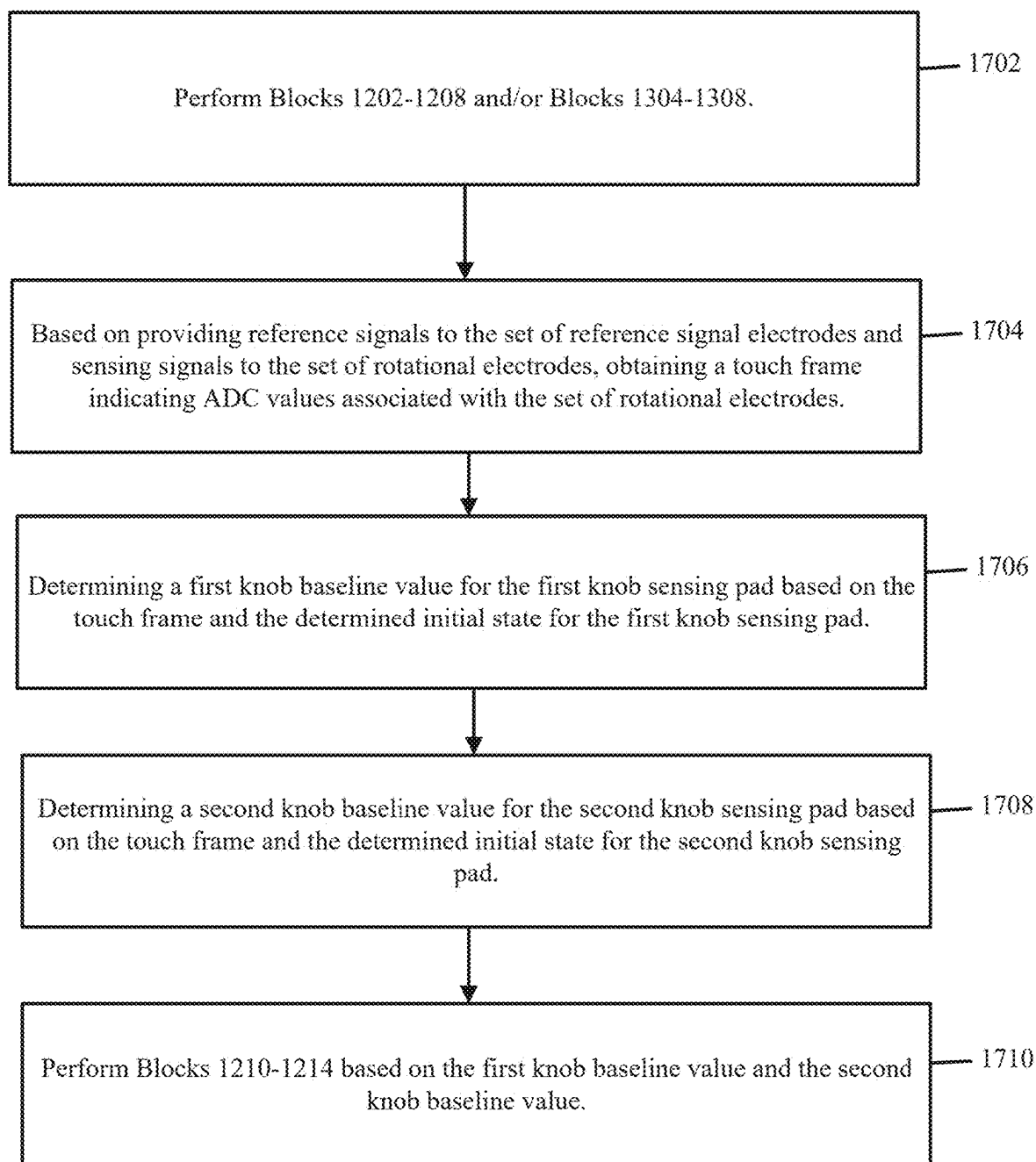


Figure 16



1700
Figure 17

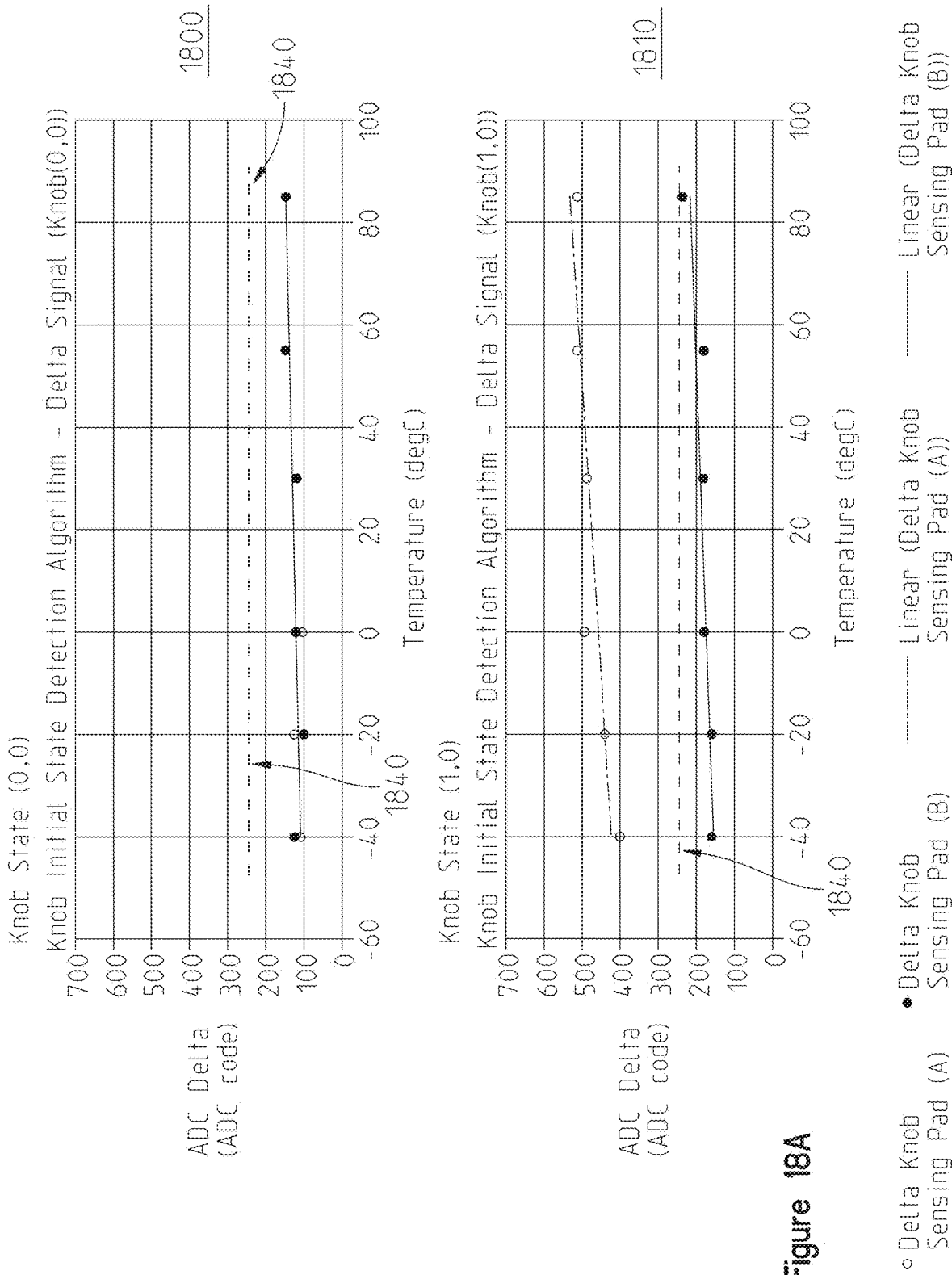


Figure 18A

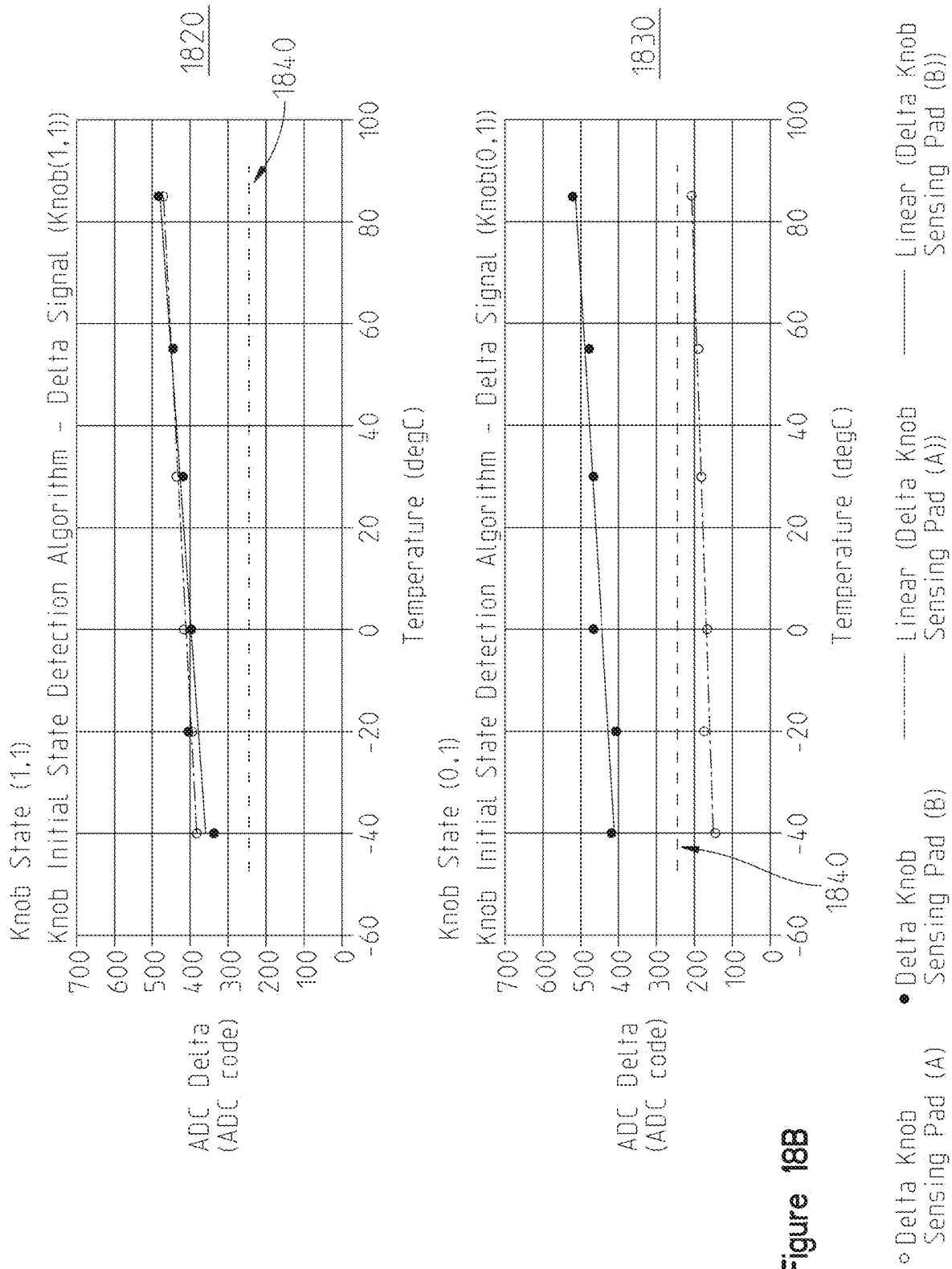


Figure 18B

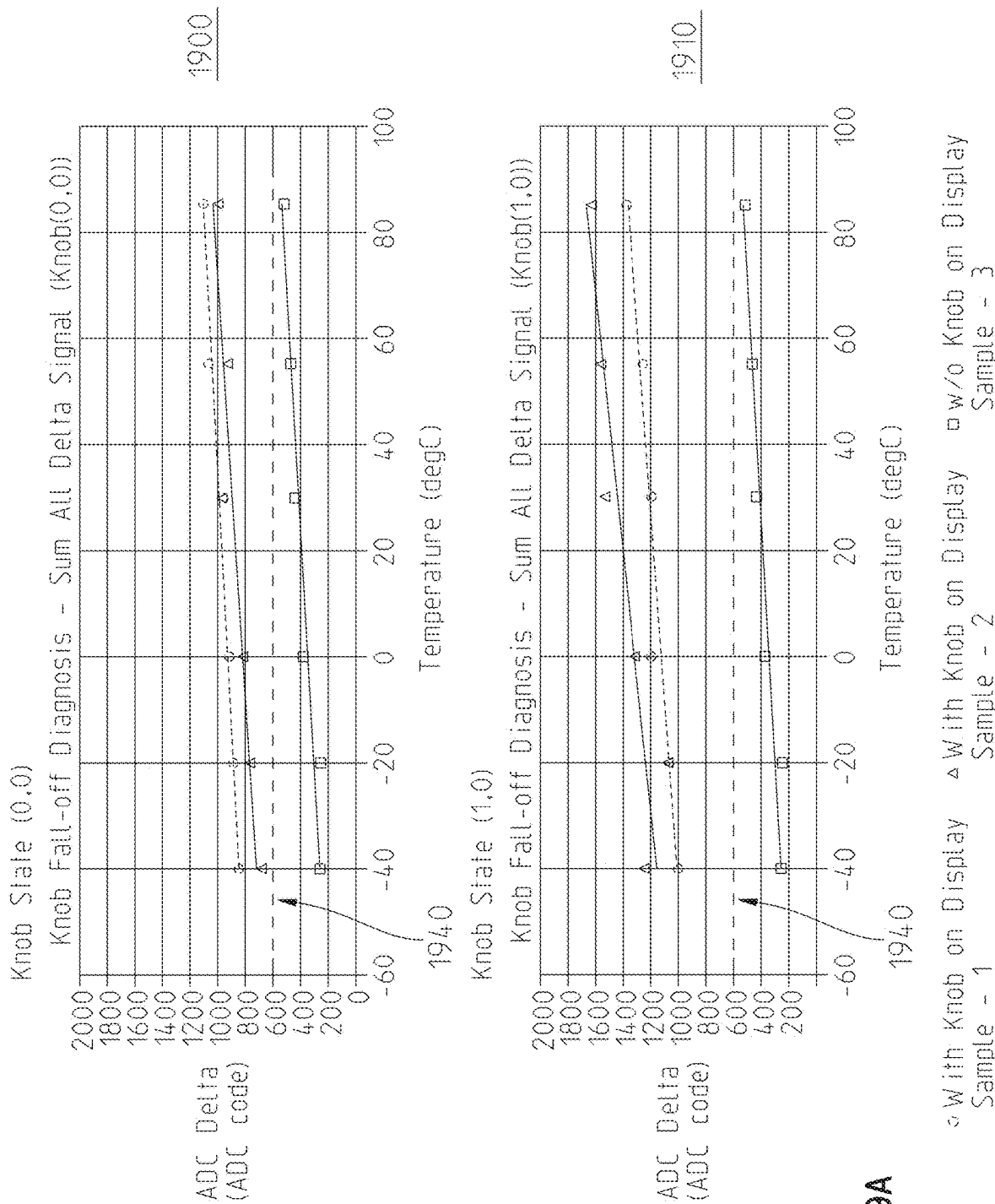


Figure 19A

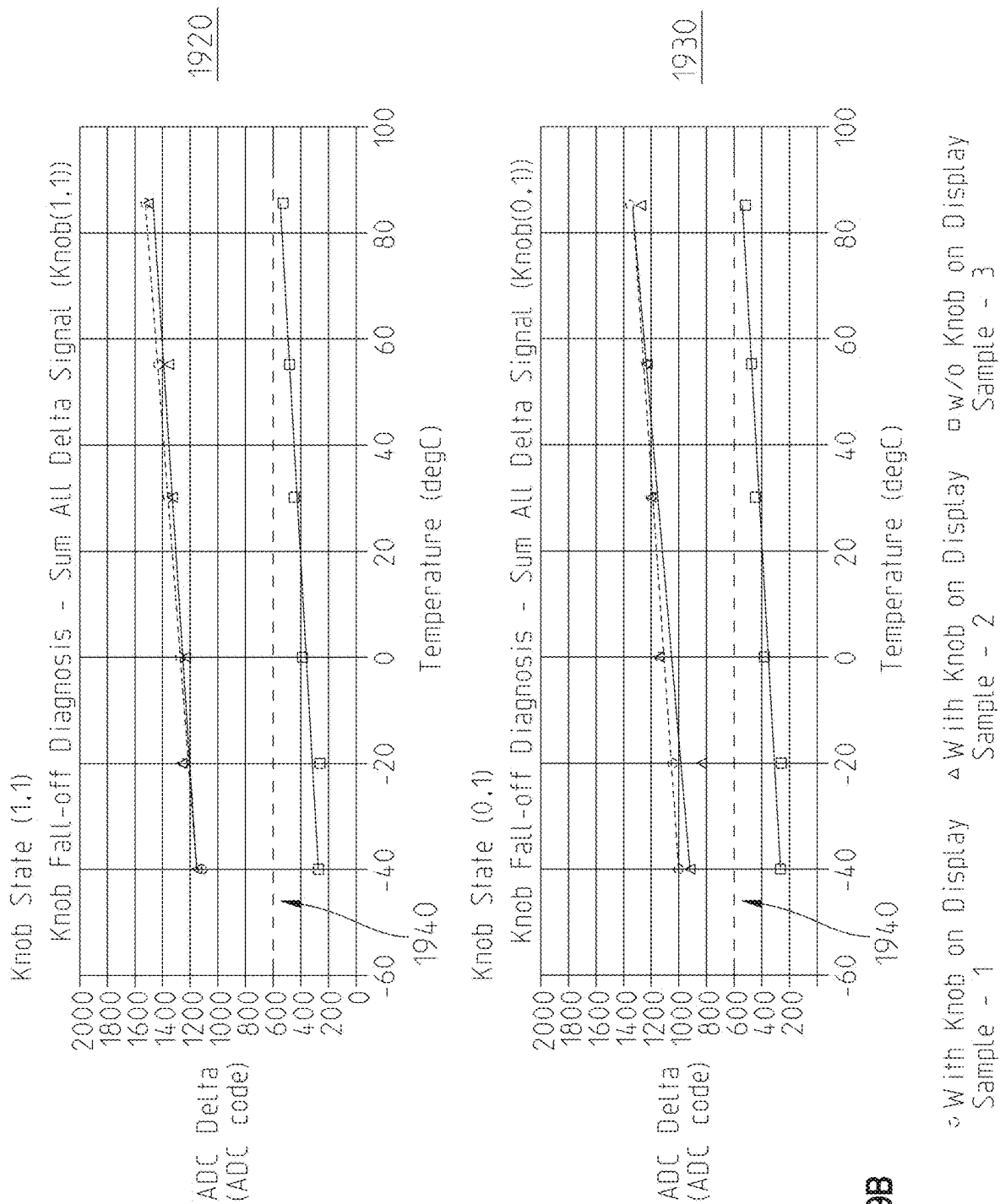


Figure 19B

CAPACITIVE KNOB SENSING SYSTEM AND METHOD TO DETECT INITIAL STATES USING A BASELINE FRAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application claims the benefit of U.S. Provisional Patent Application No. 63/557,766, filed Feb. 26, 2024, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to a rotatable knob interface.

BACKGROUND

[0003] Input devices including proximity sensor devices may be used in a variety of electronic systems. A proximity sensor device may include a sensing region, demarked by a surface, in which the proximity sensor device determines the presence, location, force and/or motion of one or more input objects. Proximity sensor devices may be used to provide interfaces for the electronic system. For example, proximity sensor devices may be used as input devices for larger computing systems, such as touchpads integrated in, or peripheral to, notebook or desktop computers. Proximity sensor devices may also often be used in smaller computing systems, such as touch screens integrated in cellular phones. Additionally, proximity sensor devices may be implemented as part of a multi-media entertainment system of an automobile. In such cases, it is useful to interface a knob to a proximity sensor device.

[0004] Traditionally, the knob interface may include a quadrature encoder rotary system that requires the rotary system to rotate by two states (e.g., two detents or resolutions) before the direction (e.g., whether the knob is being rotated clockwise or counterclockwise) is determined. However, a user may seek to adjust a setting (e.g., fan, temperature, and/or volume) by a single state. If the user rotates the knob by a single state, the system might not be capable of adjusting the setting as the direction (e.g., clockwise or counterclockwise) is unknown.

SUMMARY

[0005] This summary is provided to introduce a selection of concepts in a simplified form that are further described below. This summary is not intended to necessarily identify key features or essential features of the present disclosure. The present disclosure may include the following various aspects and embodiments.

[0006] In an exemplary embodiment, the present disclosure provides a method for determining a rotational direction of a rotatable knob interface. The method comprises: based on providing reference signals to a set of alternative electrodes, obtaining, by a processing system, a baseline analog-to-digital (ADC) frame indicating ADC values associated with a set of rotational electrodes; based on providing the reference signals to a first subset of a set of reference signal electrodes, obtaining, by the processing system, a first ADC frame indicating ADC values associated with one or more first rotational electrodes from the set of rotational electrodes; based on providing the reference signals to a second subset of the set of reference signal electrodes, obtaining, by

the processing system, a second ADC frame indicating ADC values associated with one or more second rotational electrodes from the set of rotational electrodes; determining, by the processing system, an initial state of the rotatable knob interface using the baseline ADC frame, the first ADC frame, and the second ADC frame; based on the rotatable knob interface being rotated from the initial state to a first rotational state, obtaining, by the processing system and using the set of rotational electrodes, first rotational state resulting signals indicating the first rotational state of the rotatable knob interface; determining, by the processing system, the rotational direction of the rotatable knob interface using the determined initial state of the rotatable knob interface and the first rotational state resulting signals; and adjusting, by the processing system, a setting using the determined rotational direction.

[0007] In another exemplary embodiment, the present disclosure provides a processing system for determining a rotational direction of a rotatable knob interface, comprising: one or more processors; and a non-transitory computer-readable medium having processor-executable instructions stored thereon. The processor-executable instructions, when executed by the one or more processors, facilitating performance of the following: based on providing reference signals to a set of alternative electrodes, obtaining a baseline analog-to-digital (ADC) frame indicating ADC values associated with a set of rotational electrodes; based on providing the reference signals to a first subset of a set of reference signal electrodes, obtaining a first ADC frame indicating ADC values associated with one or more first rotational electrodes from the set of rotational electrodes; based on providing the reference signals to a second subset of the set of reference signal electrodes, obtaining a second ADC frame indicating ADC values associated with one or more second rotational electrodes from the set of rotational electrodes; determining an initial state of the rotatable knob interface using the baseline ADC frame, the first ADC frame, and the second ADC frame; based on the rotatable knob interface being rotated from the initial state to a first rotational state, obtaining, using the set of rotational electrodes, first rotational state resulting signals indicating the first rotational state of the rotatable knob interface; determining the rotational direction of the rotatable knob interface using the determined initial state of the rotatable knob interface and the first rotational state resulting signals; and adjusting a setting using the determined rotational direction.

[0008] In yet another exemplary embodiment, the present disclosure provides a non-transitory computer-readable medium having processor-executable instructions stored thereon. The processor-executable instructions, when executed, facilitating performance of the following: based on providing reference signals to a set of alternative electrodes, obtaining a baseline analog-to-digital (ADC) frame indicating ADC values associated with a set of rotational electrodes; based on providing the reference signals to a first subset of a set of reference signal electrodes, obtaining a first ADC frame indicating ADC values associated with one or more first rotational electrodes from the set of rotational electrodes; based on providing the reference signals to a second subset of the set of reference signal electrodes, obtaining a second ADC frame indicating ADC values associated with one or more second rotational electrodes from the set of rotational electrodes; determining an initial state of the rotatable knob interface using the baseline ADC

frame, the first ADC frame, and the second ADC frame; based on the rotatable knob interface being rotated from the initial state to a first rotational state, obtaining, using the set of rotational electrodes, first rotational state resulting signals indicating the first rotational state of the rotatable knob interface; determining the rotational direction of the rotatable knob interface using the determined initial state of the rotatable knob interface and the first rotational state resulting signals; and adjusting a setting using the determined rotational direction.

[0009] Further features and aspects are described in additional detail below with reference to the FIGs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 illustrates an example input device with a rotatable knob interface according to one or more examples of the present disclosure.

[0011] FIG. 2 illustrates a cross-sectional side view of an example rotatable knob interface according to one or more examples of the present disclosure.

[0012] FIG. 3 illustrates an exploded view of the example rotatable knob interface of FIG. 2 according to one or more examples of the present disclosure.

[0013] FIG. 4A illustrates an underside view of the fixed base of an example rotatable knob interface as shown in FIG. 3 with a first set of reference electrodes and two sets of sensing electrodes according to one or more examples of the present disclosure.

[0014] FIG. 4B illustrates an example portion of an input device with an electrode grid with two sets of electrodes according to one or more examples of the present disclosure.

[0015] FIG. 4C illustrates the fixed base of an example rotatable knob interface of FIG. 4A as positioned over the example sensor grid of FIG. 4B according to one or more examples of the present disclosure.

[0016] FIG. 5 illustrates a perspective top view, a bottom view, and another top view of the example fixed base of FIGS. 3 and 4A through 4C according to one or more examples of the present disclosure.

[0017] FIG. 6A illustrates exploded and collapsed views of the example fixed base and example plastic bearings shown in FIG. 3 according to one or more examples of the present disclosure.

[0018] FIG. 6B illustrates the respective exploded and collapsed views shown in FIG. 6A, with the addition of the example rotary wheel of FIG. 3 provided on top of an example flat ring-shaped bearing according to one or more examples of the present disclosure.

[0019] FIG. 7A illustrates a detailed bottom view of the rotary wheel of FIG. 3 according to one or more examples of the present disclosure.

[0020] FIG. 7B illustrates a detailed top view of the rotary wheel of FIG. 7A according to one or more examples of the present disclosure.

[0021] FIG. 8 depicts the top view of the example fixed base and the bottom view of the example rotary wheel, as shown in FIGS. 5 and 7A, respectively, and capacitive coupling between them according to one or more examples of the present disclosure.

[0022] FIG. 9 illustrates example channel assignment of the outer region conductive pads of the top of the fixed base according to one or more examples of the present disclosure.

[0023] FIG. 10 illustrates example digitized quadrature encoder signals that may be generated by interaction of an

example rotary wheel with an upper surface of the example fixed base of FIG. 9 according to one or more examples of the present disclosure.

[0024] FIG. 11A illustrates four exemplary coupling states between the “A” and “B” designated conductive pads of the top of the fixed base of FIG. 9 and the bottom of the rotary wheel according to one or more examples of the present disclosure.

[0025] FIG. 11B illustrates an example distance between the two outer region conductive pads of the top of the fixed base of FIG. 9 according to one or more examples of the present disclosure.

[0026] FIG. 12 is a flowchart of an exemplary process for determining a direction of rotation of the rotatable knob interface according to one or more examples of the present disclosure.

[0027] FIG. 13 is another flowchart of an exemplary process for determining a direction of rotation of the rotatable knob interface according to one or more examples of the present disclosure.

[0028] FIGS. 14-16 show analog-to-digital (ADC) frames indicating ADC values according to one or more examples of the present disclosure.

[0029] FIG. 17 is yet another flowchart of an exemplary process for determining a direction of rotation of the rotatable knob interface according to one or more examples of the present disclosure.

[0030] FIGS. 18A and 18B show knob initial state detection graphical representations for different knob states according to one or more examples of the present disclosure.

[0031] FIGS. 19A and 19B show knob fall-off diagnosis graphical representations for different knob states according to one or more examples of the present disclosure.

DETAILED DESCRIPTION

[0032] The following detailed description is exemplary in nature and is not intended to limit the disclosure or the application and uses of the disclosure. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding background, summary and brief description of the drawings, or the following detailed description.

[0033] In the following detailed description, numerous specific details are set forth in order to provide a more thorough understanding of the disclosed technology. However, it will be apparent to one of ordinary skill in the art that the disclosed technology may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.

[0034] Various examples of the present disclosure provide input devices and methods that determine a direction of the knob that is being rotated by a user after a single state (e.g., a single detent or a single resolution) instead of two or more states. For instance, the present disclosure determines the initial states of the knob sensing pads, which are used for detecting the rotation of the knob. The knob sensing pads may detect values of the sensing touch pixels underneath the knob sensing pads. These values may change depending on if the knob conductive pads are grounded. After determining the initial states of the knob sensing pads, the present disclosure may determine the initial states of the knob, and use the initial states of the knob to determine the direction of rotation after a single resolution.

[0035] For instance, traditionally, various applications (e.g., automotive applications) may use a quadrature encoder signal to detect rotation. There are four unique states for two different channels that are offset in position by a specified amount. As the user rotates the knob, a step signal is generated for each channel (e.g., “0” or Low and “1” or High). A direction of rotation may be determined by analyzing the signals from each channel. However, as mentioned above, traditional systems require the knob to be rotated by two states (e.g., two resolutions or two detents) before the system can determine the direction of rotation.

[0036] As such, the present disclosure determines the direction of rotation for the knob after the knob has been rotated by a single state (e.g., a single resolution or detent). For instance, by determining the initial state of the knob upon start-up, the present disclosure determines the direction of rotation of the knob after a single state of rotation, which reduces the number of detents required to identify direction from two states to a single state. Accordingly, among other advantages, because the input device of the present disclosure may determine a rotational direction of the knob after the knob has been rotated a single state, the present disclosure provides for a better user experience. For instance, the user may seek to adjust a setting (e.g., a volume or a fan speed) by a single state (e.g., reduce the volume from “15” to “14” or reduce the fan speed from “5” to “4”). However, for traditional systems, if the user rotates the knob by a single state, traditional systems would fail to determine whether to increase or decrease the setting (e.g., whether to reduce the volume from “15” to “14” or to increase the volume from “15” to “16”). As such, aspects of the present disclosure may determine the direction of rotation for the knob after the knob has been rotated by a single state, which provides for a better user experience. Example knob interfaces are described in U.S. Pat. No. 10,921,913, titled “Rotatable knob interface,” and U.S. patent application Ser. No. 17/821,866, titled “Capacitive knob sensing system and method to detect initial states,” which are incorporated by reference in their entirety herein.

[0037] FIG. 1 is a block diagram depicting an electronic device (e.g., an input device) according to one or more examples of the present disclosure. The electronic device 100 may be configured to provide input to an electronic system, and/or to update one or more devices. As used herein, the term “electronic system” (or “electronic device” or “input device”) broadly refers to any system capable of electronically processing information. Some non-limiting examples of electronic systems include personal computers of all sizes and shapes, such as desktop computers, laptop computers, netbook computers, tablets, web browsers, e-book readers, and personal digital assistants (PDAs). Additional examples of electronic systems include composite input devices, such as physical keyboards that include the electronic device 100 and separate joysticks or key switches. Further examples of electronic systems include peripherals such as data input devices (including remote controls and mice), and data output devices (including display screens and printers). Other examples include remote terminals, kiosks, and video game machines (e.g., video game consoles, portable gaming devices, and the like). Other examples include communication devices (including cellular phones, such as smart phones), and media devices (including recorders, editors, and players such as televisions, set-top boxes, music players, digital photo frames, and

digital cameras). Additionally, the electronic system could be a host or a slave to the input device. In other embodiments, the electronics system may be part of an automobile, and the electronic device 100 represents one or more sensing devices of the automobile. In some instances, an automobile may include multiple electronic devices 100, where each electronic device 100 may be configured differently from the other electronic devices.

[0038] The electronic device 100 may be implemented as a physical part of the electronic system, or may be physically separate from the electronic system. As appropriate, the electronic device 100 may communicate with parts of the electronic system using any one or more of the following: buses, networks, and other wired or wireless interconnections. Example communication protocols include Inter-Integrated Circuit (I²C), Serial Peripheral Interface (SPI), Personal System/2 (PS/2), Universal Serial Bus (USB), Bluetooth®, Radio Frequency (RF), and Infrared Data Association (IrDA) communication protocols.

[0039] In some variations, the electronic device 100 may utilize any combination of sensor components and sensing technologies to detect user input. For example, as show in FIG. 1, the electronic device 100 includes one or more electrodes 125 (e.g., sensor electrodes and/or drive electrodes) that may be driven to detect objects or update one or more devices. In some instances, the electrodes 125 are sensor electrodes of a capacitive sensing device. In such instances, the electrodes 125 include one or more common voltage electrodes. In other instances, the electrodes 125 are electrodes of an image sensing device, radar sensing device, and/or ultrasonic sensing device. Further yet, the electrodes 125 may be display electrodes of a display device. In some examples, the electrodes 125 of the electronic device 100 are comprised of the common electrodes and have a common shape. Some of the examples described herein include a matrix sensor input device. As described in detail below, electronic device 100 may be provided with a rotatable knob interface 150, which may interact with some or all of electrodes 125.

[0040] The electrodes 125 may have any shape, size and/or orientation. For example, the electrodes 125 may be arranged in a two-dimensional array as illustrated in FIG. 1. Each of the electrodes 125 may be substantially rectangular in shape. In other examples, the electrodes 125 may have other shapes. Further, each of the electrodes 125 may have the same shape and/or size. In other examples, at least one sensor electrode may have a different shape and/or size than another sensor electrode. In some variations, the electrodes 125 may be diamond shaped, have interdigitated fingers to increase field coupling, and/or have floating cut-outs inside to reduce stray capacitance to nearby electrical conductors.

[0041] In some instances, capacitive implementations may utilize “self-capacitance” (or “absolute capacitance”) sensing methods based on changes in the capacitive coupling between sensor electrodes and an input object. In various examples, an input object 145 near the sensor electrodes, such as, for example, finger or stylus, alters the electric field near the electrodes 125, thus changing the measured capacitive coupling. In some instances, an absolute capacitance sensing method operates by modulating sensor electrodes with respect to a reference voltage (e.g., system ground), and by detecting the capacitive coupling between the sensor electrodes and input objects.

[0042] In some variations, capacitive implementations may utilize “mutual capacitance” (or “transcapacitance”) sensing methods based on changes in the capacitive coupling between sensor electrodes. In some instances, an input object near the sensor electrodes alters the electric field between the sensor electrodes, thus changing the measured capacitive coupling. In some examples, a transcapacitive sensing method operates by detecting the capacitive coupling between one or more transmitter sensor electrodes (also “transmitter electrodes” or “transmitters”) and one or more receiver sensor electrodes (also “receiver electrodes” or “receivers”). Transmitter sensor electrodes may be modulated relative to a reference voltage (e.g., system ground) to transmit transmitter signals. Receiver sensor electrodes may be held substantially constant relative to the reference voltage, or modulated with reference to the transmitter sensor electrodes to facilitate receipt of resulting signals. A resulting signal may comprise effect(s) corresponding to one or more transmitter signals, and/or to one or more sources of environmental interference (e.g., other electromagnetic signals). Sensor electrodes may be dedicated transmitters or receivers, or may be configured to both transmit and receive.

[0043] Capacitive sensing devices may be used for detecting input objects in proximity to and/or touching input devices. Additionally, and/or alternatively, capacitive sensing devices may be used to sense features of a fingerprint. Additionally, and/or alternatively, as in the example of FIG. 1, capacitive sensing devices may be provided with a rotatable knob interface that is electrically coupled to the capacitive sensing device, and may be used to sense the rotary position of the rotary knob. In some examples that include the rotatable knob interface, the rotatable knob interface may have a home position and a compressed position, and the sensing device may also be used to determine when the rotatable knob is in the home position, and when it is in the compressed position, based on a change in capacitive coupling of one or more of electrodes 125.

[0044] The electronic device 100 further includes a processing system 110. The processing system 110 is configured to operate hardware of the electronic device 100. The processing system 110 comprises a driver module 140 (e.g., a driver device), which may include a signal generator. In some examples, the driver module 140 generates sensing signals with which to drive the electrodes 125. In some instances, the processing system 110 comprises parts of or all of one or more integrated circuits (ICs) and/or other circuitry components

[0045] In some variations, the processing system 110 also comprises electronically-readable instructions, such as firmware code, software code, and/or the like. In some instances, components of the processing system 110 are located together, such as, for example, near sensing element(s) of the electronic device 100. In other instances, components of processing system 110 are physically separate with one or more components in proximity to the sensing element(s) of electronic device 100, and one or more components elsewhere. For example, the electronic device 100 may be a peripheral coupled to a desktop computer, and the processing system 110 may comprise software configured to run on a central processing unit (CPU) of the desktop computer and one or more integrated circuits (ICs) (perhaps with associated firmware) separate from the CPU. As another example, the electronic device 100 may be physically integrated in a phone, and the processing system 110 may comprise circuits

and firmware that are part of a main processor of the phone. Further yet, the processing system 110 may be implemented within an automobile, and the processing system 110 may comprise circuits and firmware that are part of one or more of the electronic control units (ECUs) of the automobile. In some examples, the processing system 110 is dedicated to implementing the electronic device 100. In other examples, the processing system 110 also performs other functions, such as operating display screens, driving haptic actuators, and/or performing other functions.

[0046] The processing system 110 may be implemented as one or more modules (e.g., devices) that operate different functions of the processing system 110 (e.g., driver module 140, or determination module 141). Each module may comprise circuitry that is a part of the processing system 110, firmware, software, or a combination thereof. In various instances, different combinations of modules may be used. Example modules include hardware operation modules for operating hardware such as sensor electrodes and display screens, data processing modules for processing data such as sensor signals and positional information, and reporting modules for reporting information. Further example modules include sensor operation modules configured to operate sensing element(s) to detect input, identification modules configured to identify gestures such as mode changing gestures, and mode changing modules for changing operation modes. In some instances, the electronic device 100 may be implemented as a chip, or as one or more chips. In some examples, the electronic device 100 may comprise a controller, or a portion of a controller, of electronic device 100.

[0047] In some instances, a display driver (e.g., driver module 140) may be configured for both display updating and input sensing, and may, for example, be referred to as including touch and display driver integration (TDDI) technology. In such instances, the driver module 140 may be implemented as a TDDI chip, or a portion of a TDDI chip. In some examples, the electronic device may include matrix sensor and may also include TDDI technology.

[0048] In some variations, the processing system 110 further includes a determination module 141 (e.g., a determination device). The determination module 141 may be configured to determine changes in a capacitive coupling between each modulated sensor electrode and an input object, such as input objects 145, from the resulting signals. In some instances, all of the electrodes 125 may be simultaneously operated for absolute capacitive sensing, such that a different resulting signal is simultaneously received from each of the electrodes or a common resulting signal from two or more electrodes. In other instances, some of the electrodes 125 may be operated for absolute capacitive sensing during a first period and others of the electrodes 125 may be operated for absolute capacitive sensing during a second period that is non-overlapping with the first period.

[0049] In some variations, the processing system 110 responds to user input (or lack of user input) directly by causing one or more actions. Example actions include changing operation modes, as well as graphic user interface (GUI) actions such as cursor movement, selection, menu navigation, and other functions. In some instances, the processing system 110 provides information about the input (or lack of input) to some part of the electronic system (e.g., to a central processing system of the electronic system that is separate from the processing system 110, if such a

separate central processing system exists). In some examples, some part of the electronic system processes information received from the processing system 110 to act on user input, such as to facilitate a full range of actions, including mode changing actions and GUI actions. Further, in some examples, the processing system 110 is configured to identify one or more objects, and the distance to these objects. In some instances, the processing system 110 is configured to identify (determine) one or more rotational changes of the rotatable knob interface 150, or one or more changes of state of the rotatable knob interface 150, or both, and map those changes to desired actions. Additionally, and/or alternatively, the processing system 110 is configured to determine an initial state of the rotatable knob interface 150. Using the initial state of the rotatable knob interface 150, the processing system 110 is configured to determine a direction of rotation of the rotatable knob interface (e.g., clockwise or counterclockwise) after one state change of the rotatable knob interface 150 (e.g., after the user rotates the rotatable knob interface 150 by a single resolution or a single detent).

[0050] For example, in some instances, the processing system 110 operates electrodes 125 to produce electrical signals (resulting signals) indicative of input (or lack of input) in a sensing region. The processing system 110 may perform any appropriate amount of processing on the electrical signals in producing the information provided to the electronic system. For example, the processing system 110 may digitize analog electrical signals obtained from the electrodes 125. As another example, the processing system 110 may perform filtering or other signal conditioning, or, as yet another example, the processing system 110 may subtract or otherwise account for a baseline, such that the information reflects a difference between the electrical signals and the baseline. As yet further examples, the processing system 110 may determine positional information, recognize inputs as commands, recognize handwriting, recognize fingerprint information, distance to a target object, and the like.

[0051] It should be understood that while many examples of the disclosure are described in the context of a fully functioning apparatus, the mechanisms of the present disclosure are capable of being distributed as a program product (e.g., software) in a variety of forms. For example, the mechanisms of the present disclosure may be implemented and distributed as a software program on information bearing media that are readable by electronic processors (e.g., non-transitory computer-readable and/or recordable/writable information bearing media readable by the processing system 110). Additionally, and/or alternatively, examples of the present disclosure apply equally regardless of the particular type of medium used to carry out the distribution. Examples of non-transitory, electronically readable media include various discs, memory sticks, memory cards, memory modules, and the like. Electronically readable media may be based on flash, optical, magnetic, holographic, or any other storage technology.

[0052] In some instances, the processing system 110 is configured to generate a voltage signal to drive the electrodes 125 during a display update interval and an input sensing interval, respectively. In such instances, the voltage signal generated to drive the electrodes 125 during a display update interval is a substantially constant, or fixed voltage, and the voltage signal generated to drive the electrodes 125

during an input sensing interval may be referred to as a sensing signal, having a waveform with a periodically variable voltage. In some examples, the value of a voltage signal to drive the electrodes 125 during a display update interval may be predetermined. For example, the voltage value may be provided by a manufacturer of electronic device 100 and/or the electrodes 125, and may be device-specific to electronic device 100.

[0053] In some instances, the driver module 140 comprises circuitry configured to provide the sensing signal. For example, the driver module circuitry may include an oscillator, one or more current conveyers and/or a digital signal generator circuit. In some examples, the circuitry of the driver module 140 generates the voltage signal based on a clock signal, the output of the oscillator and the parameters discussed above.

[0054] As noted above, in some instances, the driver module 140 generates a signal to drive the electrodes 125 during each of the display update periods and input sensing update periods. In such instances, an input sensing update period is provided in between two display update periods, and is of a shorter duration than a display update period. In such instances, there are several display update periods and input sensing update periods per display frame. In some examples, by acquiring the resulting signals over successive input sensing periods, the rotation of the rotatable knob interface 150, as well as whether it is in its home state or compressed state, may be tracked.

[0055] As noted above, in some variations, an additional input apparatus may be provided on top of the display panel 120 of the electronic device 100, such as, for example, the rotatable knob interface 150, and may be electrically coupled to some or all of electrodes 125 that are positioned near or below it. In some instances, the additional apparatus may provide alternate ways for a user to provide input to electronic device 100 other than touching, or hovering near, a display screen with a finger or stylus. In the depicted example of FIG. 1, the rotatable knob interface 150 is mounted onto the display panel 120, and may have a full (as shown in FIG. 1) or partial overlap with the display panel 120. As noted, in one or more examples, the rotatable knob interface 150 may have a stationary base that is provided with various sets of coupling electrodes configured to couple with respective sets of electrodes of the display panel 120, such as one or more sets of electrodes that are provided with sensing signals and one or more sets of electrodes that are provided with reference signals. In some variations, the stationary base may include different conductive regions respectively connected to corresponding sets of coupling electrodes.

[0056] In some instances, the rotatable knob interface 150 also includes a rotary wheel that sits above, and rotates relative to, the stationary base. In such instances, an underside of the rotary wheel is patterned with various conductive and non-conductive regions in a peripheral region 152, configured to align with the conductive regions of the stationary base so that there are various electrical couplings between the conductive regions of the stationary base and the various conductive and non-conductive regions in the peripheral region 152 of the rotary wheel. These components are further configured such that these electrical couplings change as the rotary wheel is rotated, in such manner that by detecting the effects of the changes in the electrical couplings on resulting signals received on the display panel, the

electronic device **100** (e.g., the input device) may determine a rotation, or a change in rotation, of the rotatable knob interface **150**. In some instances, the peripheral region **152** may have numerous possible example arrangements of the conductive and non-conductive regions, and there may be various ways of having the rotary wheel and the stationary base electrically interact as the rotary wheel is rotated. Thus, alternate configurations and relative arrangements of both the conductive regions of the stationary base, and the placement of the conductive and non-conductive regions of the rotary wheel are possible, all being within the scope of this disclosure.

[0057] In some examples, the rotation imparted to the rotatable knob interface by a user, in either relative or absolute terms, may be detected by the electronic device **100**. In some instances, the rotatable knob interface **150** may also be pressed downwards by a user, and may thus have two positions: a home or “uncompressed” position, and a “compressed” position, which a user maintains by, for example, pushing down on the rotatable knob interface **150** against one or more biasing springs. In some instances, the rotatable knob interface **150** has a cover. In some variations, the rotatable knob interface may be pressed downwards so as to rest at multiple positions, and thus may have multiple states between an “uncompressed” and a “fully compressed” position. In the home position, the cover is at a greater distance above the rotary wheel than in the compressed position. In some variations, the rotary wheel may have several switches provided between the rotary wheel and the cover, and these switches may include biasing springs. In such variations, the rotatable knob interface **150** may be provided with a fourth set of coupling electrodes, which couple to electrodes of the input device that are also driven with sensing signals. In the example of FIG. 1, the fourth set of coupling electrodes is connected to an inner ring provided in the stationary base, which aligns with a similarly shaped inner ring **153** that is provided in the rotary wheel. In such examples, when a user presses down on the cover of the rotatable knob interface, so that the rotatable knob interface **150** is then in the “compressed” position, the switches close so as to connect the inner ring **153** of the rotary wheel with all of the conductive regions provided in peripheral region **152**. This serves to electrically couple the fourth set of coupling electrodes of the stationary base to the first set of coupling electrodes of the stationary base, thereby coupling a corresponding fourth set of electrodes of the display panel to a reference signal. However, when the user ceases to press down on the cover, the fourth set of coupling electrodes of the rotatable knob interface simply floats. In some instances, the direction and degree of rotation, as well as a user pressing down on, or ceasing to press down upon, the rotatable knob interface **150**, may be interpreted by processing system **110**, such as, for example, by the determination module **141**, and may be mapped to various user input actions, signals, or directives.

[0058] In some instances, a user may rotate the rotatable knob interface **150** in various ways, for example, grabbing an outer housing of the rotatable knob interface and turning it, grabbing a top of the rotatable knob interface, or a flange protruding from the side of the rotatable knob interface and turning it, or placing one or more fingertips in or on a recessed channel on an upper surface of the rotatable knob interface.

[0059] In some instances, the electronic device **100** of FIG. 1 may be provided in an automobile. For example, the

electronic device **100** may be affixed to a substantially vertical display screen provided in a central part of a dashboard. In some variations, all the electrodes not physically blocked by the rotatable knob interface **150**, whether the electrodes **125** are inside or are outside of region **155** (described below), remain active. Thus, in such variations, both touches away from the knob, and rotations of the knob, are detected and reported by the electrodes **125** at the same time.

[0060] In some examples, all other forms of user input besides those received via the rotatable knob interface **150** may be disabled on the electronic device. Thus, in such examples, the electrodes **125** are not driven during the sensing interval to perform their standard sensing functionality. As a result, if a finger or other input object **145** is moved into, or away from, its vicinity, no resulting signal is obtained, or if obtained, it is not processed. In such examples, this may be done to prevent a driver of the automobile from attempting to touch the display **120** while driving, as a safety measure, and thus to only interact with the electronic device **100** via the rotatable knob interface **150**. In such examples, the disabling of standard sensing functionality of the electrodes **125** may be implemented during specified activities of the automobile, but not during others. For example, the disabling of standard sensing functionality of the electrodes **125** may be implemented while the automobile is in actual motion, but at all other times some of the electrodes **125**, for example, those not near enough to the rotatable knob interface to interfere with signals acquired from it, may be operated to perform standard sensing, as described above.

[0061] In some instances, when all of the electrodes **125** are disabled from standard sensing, whether during actual driving of the automobile, or whether at all times, as the case may be, the only way that a driver of the automobile can provide input to the electronic device **100** is via the rotatable knob interface **150**, using a pre-defined set of rotations and/or pressings of the rotatable knob interface **150**. These motions modify a resulting signal which is received by the electronic device **100** during a sensing period, which then interprets them, for example, using determination module **141**. The resulting signal may be the same signal as the sensing signal that driver module **140** drives an electrode **125** with, after being modified by the capacitive coupling of the rotatable knob interface **150**.

[0062] In some instances, for example, only some of the electrodes **125**, in particular those that are near or beneath the rotatable knob interface **150**, are disabled from standard capacitive sensing, and the remainder of the electrodes **125** on the electronic device **100** may still be operative for standard capacitive sensing. In such instances, the electrodes that are disabled for standard capacitive sensing are those that are close enough to the rotatable knob interface **150** such that driving them with standard sensing signals may interfere with the resulting signals obtained from various sets of the electrodes **125** that are respectively electrically coupled to the coupling electrodes of the rotatable knob interface **150**. To illustrate this feature, in FIG. 1, there is shown a region **155** (e.g., a dashed line boundary). Electrodes **125** within the region **155** are in a “blackout zone” and not driven with a standard sensing signal. Rather, as described in detail below, any of the electrodes within the blackout zone that are electrically coupled to the rotatable

knob interface are driven so as to capture rotations and compressions of the rotatable knob interface, as described below.

[0063] In general, within the blackout zone, a first, second and third set of the electrodes **125** are coupled to corresponding first, second and third sets of the coupling electrodes of the stationary base of the rotatable knob interface **150**. In some instances, the first set are driven with a reference signal, and the second and third sets are driven with a sensing signal to obtain a resulting signal modified by the then extant relative rotational relationship of the stationary base and the rotary wheel of the rotatable knob interface **150**. Thus, in each of these instances, the electrodes within the region **155** (e.g., a blackout zone boundary) may be disabled from standard capacitive sensing at all times.

[0064] Furthermore, as noted above, sets of electrodes of the electronic device **100** are electrically coupled to corresponding sets of coupling electrodes of the rotatable knob interface **150**. Thus, during an input sensing period, a reference signal is supplied by the driver module **140** to a first set of the electrodes **125**, and a sensing signal is supplied to the second and third sets of the electrodes **125**. In some instances, the reference signal may be a configurable direct current (DC) output provided by the processing system **110**. In some examples, the DC signal may be a ground signal of the electronic device **100**. In some variations, a resulting signal is obtained from each of the second and third sets of the electrodes **125**, where the resulting signals is the sensing signal as modified by the rotational state of the rotatable knob interface **150**. The resulting signals are interpreted by the determination module **141** to determine a rotation of the rotatable knob interface **150**. For instance, using the resulting signals, the processing system **110** may determine a number of states (e.g., detents or resolutions) that the user has rotated the rotatable knob interface **150** as well as the direction of the rotation. Furthermore, upon start-up (e.g., any time prior to the user rotating the rotatable knob interface **150**), the processing system **110** may determine an initial state of the rotatable knob interface **150**. For instance, the processing system **110** may obtain resulting signals from the second and/or third set of electrodes. Based on comparing the resulting signals from multiple frames, the processing system **110** may determine an initial state of the rotatable knob interface **150**. Subsequently, the processing system **110** may determine a new state of the rotatable knob interface associated with the user rotating the rotatable knob interface **150**. By comparing the initial state of the rotatable knob interface **150** and the new state of the rotatable knob interface **150**, the processing system **110** may determine the direction of rotation of the rotatable knob interface **150**. Thus, the processing system **110** does not need two separate states of rotation of the rotatable knob interface **150** to determine the direction as the processing system **110** uses an initial state of the rotatable knob interface **150** (e.g., a state when the rotatable knob interface **150** has not been rotated) and one additional state of the rotatable knob interface **150** during rotation by a user to determine the direction of rotation.

[0065] Additionally, and/or alternatively, the processing system **110** may obtain additional resulting signals from additional sensing electrodes associated with the rotatable knob interface **150**. For instance, in-between the first set of electrodes and the second/third sets of electrodes, there may be an empty column of sensing pixels (e.g., the empty

column **412** and/or the empty columns to the left of the sensing electrodes **410** and **411** from FIG. 4A below) that are coupled to one or more sensing electrodes. Using the resulting signals from these sensing electrodes as well as the resulting signals from the second and/or third set of electrodes, the processing system **110** determines an initial state of the rotatable knob interface **150**. Using the initial state of the rotatable knob interface **150** and an additional state of the rotatable knob interface **150**, the processing system **110** determines the direction of rotation of the rotatable knob interface **150**. This will be explained in further detail below.

[0066] FIG. 2 illustrates a cross-sectional side view of an example rotatable knob interface according to one or more examples of the present disclosure. For instance, starting at the bottom, the rotatable knob interface includes a fixed base **231**. In some instances, the fixed base **231** does not move as a user rotates the example rotatable knob interface. Thus, in some examples, the fixed base **231** is affixed to the surface of an example input device, such as, for example, by an adhesive. In some variations, the fixed base **231** is affixed to the input device in a semi-permanent or permanent manner, and is placed thereon so as to align with a grid of electrodes provided in the input device. Provided above the fixed base **231** is a rotary wheel **230**. The rotary wheel **230** turns as a user rotates the rotatable knob interface, such as, for example, by grasping and turning cover cap **215**, as described below. At an inner side of the rotary wheel **230** is provided a vertical ring bearing **225**. The vertical ring bearing **225** is non-conductive, and may be made of plastic, for example, and may have the shape of a ring. Vertical ring bearing **225** may have a substantially tubular shape. Not shown in FIG. 2, but described below with reference to FIG. 3, is an additional substantially horizontal ring-shaped bearing upon which the rotary wheel **230** sits according to one or more examples of the present disclosure. By using both of the bearings, frictional forces between the fixed base **231**, and the rotary wheel **230** may be reduced.

[0067] Continuing with reference to FIG. 2, provided on top of rotary wheel **230** are one or more switches **220**. For example, switches **220** may be dome switches. There may be three switches **220**, and the switches may be equidistantly placed on an upper surface of rotary wheel **230**. In one or more examples, the switches are used to distinguish between two states of the rotatable knob interface such as a compressed state, in which the switches are closed, and an uncompressed state in which the switches remain open. The compression state of the rotatable knob interface is orthogonal to its internal rotational position. Thus, the rotatable knob interface may be rotated while in either a compressed or an uncompressed state (and in any position in between the two states), and that rotation may be sensed and measured. Similarly, the state of the switches as being open or closed, corresponding respectively to the rotatable knob interface being in the “home” or uncompressed state, or in the compressed state, may be detected whether or not the rotatable knob interface is rotationally stationary or being rotated.

[0068] Furthermore, the rotatable knob interface has an inner cap **210** and a cover cap **215**. In operation, a user physically interacts with cover cap **215**, for example, by grasping cover cap **215** and rotating the rotary wheel **230** relative to the fixed base **231**, or by pushing down on cover cap **215** to compress the rotatable knob interface and close the switches **220**. As shown, the inner cap **210** is attached,

by prongs 211, to a lip provided on the inner surface of vertical ring bearing 225. The cover cap 215 is attached to the inner cap 210, such that turning the cover cap 215 rotates the rotary wheel 230.

[0069] FIG. 3 shows an exploded view of the example rotatable knob interface of FIG. 2, and illustrates the upper side of various components. Beginning at the bottom, FIG. 3 shows the upper surface of the fixed base 231. The upper surface is provided with a conductive peripheral ring 235, to be coupled to a reference signal of an input device to which the rotary knob is to be attached. As shown, the upper surface also shows a conductive inner ring 232 as well as two conductive pads 237 and 238. In some instances, these three conductive regions are configured to be coupled to a sensing signal of the input device. Details of these regions, their functions, and how they interact with the input device upon which the rotary knob sits, are described in greater detail below.

[0070] FIG. 3 further shows the vertical ring bearing 225, and a horizontal ring bearing 226, configured to slide over it. In some instances, because the fixed base 231 has a smaller inner diameter than the rotary wheel 230, there is a ledge at the inner periphery of the fixed base 231 upon which the vertical ring bearing 225 may sit. The vertical ring bearing 225 is thus configured to fit inside the inner diameter of the horizontal ring bearing 226, and rest upon the inner periphery of the fixed base 231. The two bearings thus provide a physical interface between the fixed base 231 and the rotary wheel 230, as noted above, which reduces friction between them as the rotary wheel 230 is moved.

[0071] FIG. 3 further shows three switches 220 provided around the upper surface of rotary wheel 230. As noted, these switches may be dome switches, for example. Above the switches 220 is shown the inner cap 210, which is configured to fit inside the vertical ring bearing 225, and be secured to the vertical ring bearing 225 by means of three prongs 211, which, in one or more examples are also placed equidistantly around the inner vertical surface of the vertical ring bearing 225. As shown, the inner cap 210 has a substantially horizontal upper ring, and a lower hollow cylindrical shaped portion. Thus, in some instances, the outer diameter of the lower cylindrical shaped portion of the inner cap 210, is designed to fit within an inner diameter of the vertical ring bearing 225, and then clamp to the bottom surface of the vertical ring bearing 225 by the prongs 211, which slightly protrude under such bottom surface when the inner cap 210 is in the home or uncompressed position. Further, the cover cap 215 is attached to the upper ring portion of the inner cap 210.

[0072] FIG. 4A illustrates an underside view of the fixed base of an example rotatable knob interface as shown in FIG. 3 with a first set of reference electrodes and two sets of sensing electrodes according to one or more examples of the present disclosure. FIG. 4B illustrates an example portion of an input device with an electrode grid with two sets of electrodes according to one or more examples of the present disclosure. FIG. 4C illustrates the fixed base of an example rotatable knob interface of FIG. 4A as positioned over the example sensor grid of FIG. 4B according to one or more examples of the present disclosure. For instance, FIGS. 4A-4C illustrate the spatial relationships between coupling electrodes provided on the bottom surface of the fixed base 231, respectively connected to corresponding conducting

regions on the top surface of the fixed base 231, and electrodes in a grid provided in an example input device.

[0073] For example, FIG. 4A illustrates a view of the underside of the fixed base 231 of an example rotatable knob interface shown in FIG. 3, superimposed over a grid 401 of electrodes of an example input device according to one or more examples of the present disclosure. For example, the bottom or underside of fixed base 231 has three sets of electrodes. A first set of electrodes 430 (e.g., a set of knob ground electrodes, reference signal electrodes, or reference ground electrodes), shown as shaded, is a connected set of electrodes configured to receive a reference signal from the input device. Three sensing electrodes 410, 420, 411, which are grouped into the remaining two sets, are configured to receive sensing waveforms of the input device. The second set, which includes sensing electrodes 410 and 411, is configured to sense rotation of the rotatable knob interface as well as determine an initial state of the rotatable knob interface. The third set, which includes electrode 420, is configured to sense a “click” or the closing of the switches 220, for example, when a user pushes the rotatable knob interface into its compressed state. In some instances, sensing electrodes 410, 411 and 420 are designed to each overlap with, to the extent possible, a full input device electrode (e.g., a square) of grid 401. On the other hand, the first set of electrodes 430 may be designed to each overlap portions of multiple electrodes of grid 401, but not full electrodes, such that the first set of electrodes 430 only picks up the signal from the corresponding reference grid electrodes 403 (see FIG. 4B) on the grid 401 on the upper surface of the example input device, and do not pick up any parasitic capacitance from neighboring sensing electrodes. This isolation is illustrated in FIG. 4A by two features. First, there is an empty column 412 of sensing pixels to the right of sensing electrodes 410, 411 and 420 that provides a gap between the sensing electrodes 410, 411 and 420, and the first set of electrodes 430. Second, the first set of electrodes 430 (full line shading) are each recessed inwardly relative to the reference grid electrodes 403 (shaded with dotted lines) by, for example, 1.5-2 millimeters (mm). This recessing helps the first set of electrodes 430 to only pick up the reference electrode signal and much less so of the parasitic coupling of nearby sensing signals on sensing grid electrodes 402. Further, this feature also helps with tolerance alignment of the example rotatable knob interface to the input device.

[0074] FIG. 4B illustrates the example grid 401 of FIG. 4A divided into two types of electrodes, according to one or more embodiments. In general, each electrode of an input device’s grid may be selectively chosen to be driven with a sensing waveform or a reference signal, such as, for example, ground, or other reference signal. In some instances, to coordinate its grid with the electrodes of the underside of a fixed base, as shown in FIG. 4A, the input device’s grid may be arranged as shown in FIG. 4B. Thus, grid electrodes 403, shaded in FIG. 4B, may be driven by the input device with a reference signal, and grid electrodes 402 may be driven by the input device with a sensing signal. In some examples, when this scheme is implemented, there is a pairing between the underside of the fixed base 231, and the electrodes of grid 401 of an input device. This is illustrated in the superimposed view of FIG. 4C.

[0075] FIG. 4C thus illustrates the underside of fixed base 231 of FIG. 4A as positioned over the example input device

electrode grid **401** of FIG. 4B according to one or more examples of the present disclosure. As shown, the sensing electrodes **410**, **411** and **420**, configured for sensing on the rotatable knob interface, are each substantially fully aligned with grid electrodes **402**, to be driven with sensing waveforms. In some instances, they are driven with the same sensing waveforms. Similarly, the first set of electrodes **430**, configured for coupling to a reference signal of the input device, are each provided above multiple grid electrodes **403**, to be driven with a reference signal by the input device. In some instances, because the fixed base **231** is stationary, and fixed in position relative to the input device, it is first aligned to the electrodes of the input device, as shown, and then, in some examples, permanently attached to a glass surface of the input device.

[0076] FIG. 5 shows the upper surface of the fixed base according to one or more examples of the present disclosure. For instance, the top perspective view **510** illustrates the positions of input device grid electrode **402** and **403** regions relative to the top surface of fixed base **231**. As shown in the top perspective view **510**, as well as by comparing bottom surface view **520** with another top view **530**, the top surface of fixed base **231** is somewhat differently organized than its bottom surface. To fully appreciate the relative positions of conductive pads on the top and bottom surfaces, bottom surface view **520** is also shown, and, as indicated by the curved arrow **521**, a corresponding position of the top surface is also shown by the top view **530**. This top view **530** is what would be seen if the fixed base **231** as shown in bottom surface view **520** was flipped about a horizontal axis (such that right and left sides of the fixed base **231** are the same in views **520** and **530**, respectively).

[0077] The top view **530** illustrates four conductive regions, namely the inner ring **232** (used to sense whether the switches are open or closed), the two conductive pads **237** and **238** (used to sense rotation) and peripheral ring **235**. In some instances, each of these is electrically connected to a corresponding conductive region on the bottom surface of fixed base **231**. For instance, peripheral ring **235** is electrically connected to a corresponding first set of electrodes **430**, as noted above, to couple to input device electrodes driven with a reference signal; the two conductive pads **237** and **238** are respectively connected to sensing electrodes **410** and **411**; and inner ring **232**, is electrically connected to sensing electrode **420**. In some examples, as noted above, both conductive pads **237** and **238**, as well as inner ring **232** are configured to couple to input device electrodes that are driven with a sensing signal.

[0078] Thus, as shown, the top of fixed base **231** has, on its outer periphery, two small conductive pads **237** and **238** near each other, surrounded by a peripheral ring **235**. The peripheral ring **235** receives a reference signal, and the two conductive pads **237** and **238** each receive a sensing signal. The two pads are used to sense rotation. A second, thinner inner ring **232** inside of the peripheral ring **235** is configured to also receive a sensing signal to sense whether the switches are closed. The closing of the switches may also be referred to as a “click” from the sound they make when they close.

[0079] FIG. 6A illustrates an exploded view **601** and a collapsed view **603** of the example fixed base **231**, the example vertical ring bearing **225**, and the horizontal ring bearing **226** (e.g., plastic bearing) shown in FIG. 3. As is shown in the collapsed view **603**, the horizontal ring bearing **226** has a smooth surface on top of which the rotary wheel

230 can rest, and the vertical ring bearing **225** has a smooth outer cylindrical structure around which the rotary wheel **230** can turn.

[0080] FIG. 6B illustrates the respective exploded view **610** and collapsed view **603** of the example fixed base **231** and the vertical and horizontal ring bearings **225**, **226**, respectively, shown in FIG. 6A, with the addition of the example rotary wheel **230** of FIG. 3 provided on top of an example flat horizontal ring bearing **226**. As shown, the vertical ring bearing **225** has a larger height than that of the rotary wheel **230**, such that it protrudes above the rotary wheel **230**. Visible in each of exploded view **610** and collapsed view **603**, are the three sets of pads **221** provided on a top surface of the rotary wheel **230** for connection to the set of switches.

[0081] FIG. 7A illustrates a detailed bottom view of the rotary wheel of FIG. 3. For instance, as in the case of the top surface of the fixed base, there are essentially two ring shaped structures—an outer peripheral ring **701**, which comprises alternating conductive regions **710** and non-conductive regions **720**, and an inner ring which comprises a single connected inner ring second conductive region **730**. Furthermore, the ring-shaped region **702**, provided between the outer peripheral ring **701** and the inner ring second conductive region **730**, is also non-conductive. In some instances, the conductive region **710** and the second non-conductive regions **720** are used to sense rotation, and the inner ring second conductive region **730** is used to sense “click.”

[0082] FIG. 7B illustrates a detailed top view of the example rotary wheel of FIG. 3. The view of FIG. 7B corresponds to the view of the top surface of rotary wheel **230** shown in FIG. 6B that illustrates three sets of pads **221**, which each respectively connect to a switch. As noted above, the switches may be dome switches, for example. However, the top view of FIG. 7B is drawn transparently, to show the underlying conducting rings to which each set of pads **221** is respectively coupled, as well as the other conductive regions on the bottom and top surfaces of the fixed base, previously described. These include, as shown here via the transparency, and as shown in FIG. 4A, on the bottom surface of fixed base **231**, sensing electrodes **410**, **420** and **411** and the first set of electrodes **430** that is coupled to a reference signal of the input device; and on the top surface of fixed base **231**, a portion of peripheral ring **235**, and conductive pads **237** and **238**.

[0083] In some instances, the conductive regions **710**, as well as conductive pads **237** and **238**, and peripheral ring **235**, may be made of known conductors, such as, for example, copper, silver, gold, aluminum, or other conductors, or, for example, various alloys of any of those, with each other, or with different elements or compounds. Similarly, in some examples, non-conductive regions **720** may be regions of a printed circuit board or substrate on which no metal is deposited, and thus be made of epoxy plastic and fiberglass, for example, or, for example, non-conductive regions **720** may be formed by depositing an insulating layer such as, for example, a silicon dioxide (SiO₂) layer.

[0084] As shown in FIG. 7B, there are two ring shaped conductive regions, namely the outer ring region **712** and the inner ring region **732**, for example, provided just under the surface of the top side of the rotary wheel **230**. The outer ring region **712** is electrically connected to each of the conductive regions **710** on the bottom side of the rotary

wheel, as shown in FIG. 7A. Similarly, the inner ring region 732, provided on the inner periphery of the top side of the rotary wheel 230, is electrically connected to the inner ring second conductive region 730 on the bottom side of the rotary wheel, also shown in FIG. 7A. Additionally, in the depicted example of FIG. 7B, while the positions of the three sets of pads 221 to which the three switches are to be connected are shown, the switches that are to respectively connect to them are not shown. Thus, when the switches are closed, by a user pushing down on the cover cap 215 (shown in FIGS. 2 and 3) until the switches make a clicking sound or equivalent, the inner portion of each pad is electrically connected to the outer portion of each pad, which causes the outer ring region 712 and the inner ring region 732 to be electrically connected. This also causes, with reference to FIG. 7A, all of the respective conductive regions 710 to be connected to the inner ring second conductive region 730. It is noted that there may be more or less switches, and corresponding sets of switch pads to which they connect, in alternate embodiments. In some instances, the pads 221 may be placed equidistantly around the rotary wheel 230. In some variations, the switches may have more than two states, and thus have more positions than “compressed” or closed, and “uncompressed” or open.”

[0085] Regarding the top and bottom surfaces of the fixed base 231 and the rotary wheel 230, FIG. 8 illustrates the electrical coupling between the top surface of the fixed base 231 and the bottom surface of rotary wheel 230, which, in one or more embodiments, face each other in the assembled rotatable knob interface, when the rotary wheel 230 sits above the fixed base 231. With reference thereto, dashed arrow 801 depicts the electrical coupling between inner ring 232 of the top surface of the example fixed base 231, and the inner ring second conductive region 730 of the bottom surface of the example rotary wheel 230. Additionally, dashed arrow 802 depicts the electrical coupling between peripheral ring 235 of the top surface of the example fixed base 231, which includes conductive pads 237 and 238, and the various conductive regions 710 of the outer peripheral ring 701 of the bottom surface of the example rotary wheel 230. As noted above, the non-conductive regions 720 of the outer peripheral ring 701, of the bottom surface of the rotary wheel 230, are non-conductive, as shown, as is the non-conductive divider ring-shaped region 702, which is provided between the outer peripheral ring 701 and the inner ring second conductive region 730. In some examples, the respective pairs of regions indicated by the dashed arrows 801 and 802 are capacitively coupled, given the non-conductive horizontal ring bearing 226 (e.g., made of plastic or the like) that sits between the two surfaces, as described above with reference to FIG. 6A.

[0086] As shown in FIG. 8, when the rotary wheel 230 is positioned above the fixed base 231 (with the horizontal bearing between them), there may be various electrical couplings between their respective peripheral ring regions. While the peripheral ring 235, which is coupled to a reference signal of the input device via the first set of electrodes 430, may be capacitively coupled to a number of conductive regions 710 of the rotary wheel underside, whether one or both of the conductive pads 237, 238 are coupled to conductive regions 710 of the rotary wheel underside depends upon the relative rotational position of the rotary wheel 230 and the fixed base 231.

[0087] In some examples, in order to sense rotation, the two conductive pads 237 and 238 on the top surface of fixed base 231 may be coupled to electrodes on the surface of the input device that are respectively driven with sensing signals. As noted above with reference to FIG. 4A, the conductive pads 237 and 238 on the top surface of fixed base 231 are respectively electrically connected with the sensing electrodes 410 and 411 provided on the bottom surface of the fixed base 231. In turn, the sensing electrodes 410 and 411 are coupled to corresponding input device electrodes that are driven with sensing signals, as shown, for example, in FIG. 4C. In some examples, by driving the input device electrodes that are respectively coupled to the fixed base sensing electrodes 410 and 411 with sensing signals, different resulting signals are received by those input device electrodes as a function of the capacitive coupling of each of the two conductive pads 237 and 238 on the top surface of fixed base 231 with the array of conductive regions 710 and non-conductive regions 720 on the bottom surface of the rotary wheel 230.

[0088] FIG. 9 is a small arcuate portion of the peripheral ring of the top surface of the fixed base. The portion shown corresponds to the portion of image shown in FIG. 7B that includes conductive pads 237 and 238. To distinguish the signals coupled to each conductive pad, with reference to FIG. 9, in some examples, the conductive pad 237 is assigned to channel A and conductive pad 238 is assigned to channel B. For convenience, for example, the conductive pad 237 may be referred to herein as the “channel A pad”, and the conductive pad 238 may be referred to as the “channel B pad.” By measuring resulting signals received by the electrodes (e.g., sensing electrodes 410 and 411 as shown in FIGS. 4A, 4C, and 5) on the input device that are respectively coupled to each of conductive pads 237 and 238 at different points in time, the processing system 110 may determine the direction of the rotation. Also shown in FIG. 9 is the peripheral ring 235 (which is coupled to the first set of electrodes 430 of the input device, and thus to the reference signal that drives them), and the conductive inner ring 232 that is used to sense “click” of the switches closing.

[0089] FIG. 10 illustrates example digitized quadrature encoder signals that may be generated by interaction of an example rotary wheel with an upper surface of the example fixed base of FIG. 9 according to one or more examples of the present disclosure. For instance, FIG. 10 illustrates examples of digitized quadrature encoder signals that may be generated by the interaction of the conductive pads 237 and 238 of the example fixed base having the example channel assignments shown in FIG. 9, with the alternating conductive regions 710 and non-conductive regions 720 of the outer peripheral ring on the bottom surface of rotary wheel 230, as rotary wheel 230 is rotated by the user. The generated signals have one sequence for clockwise rotation (e.g., a clockwise rotation sequence 1010), and another sequence for anticlockwise rotation (e.g., an anticlockwise rotation sequence 1020). The relative rotation may be determined in firmware by comparing successive sequences or states. As shown, the respective signals used for channels A and B are identical, but are shifted 90 degrees in phase. These signals may be better understood with reference to all of the possible overlap states between conductive pads 237 and 238 of fixed base 231 with the underside pattern of rotary wheel 230, as is illustrated in FIG. 11A. For instance,

the four data points 1030 of the anticlockwise rotation sequence 1020 are shown in and described in FIG. 11A.

[0090] FIG. 11A illustrates four exemplary coupling states between the “A” and “B” designated conductive pads of the top of the fixed base of FIG. 9 and the bottom of the rotary wheel according to one or more examples of the present disclosure. For instance, FIG. 11A illustrates four possible coupling states, 1110 through 1140, of the “A” and “B” designated conductive pads 237 and 238 on the top of the fixed base 231 of FIG. 9, with the pattern of alternating conductive regions 710 and non-conductive regions 720 of the outer peripheral ring on the bottom of the rotary wheel 230. In FIG. 11A, only a small portion of the peripheral ring 235 of the fixed base, near where the conductive pads 237 and 238 are provided, is shown. The relative positions of the conductive pads 237 and 238 to the underside of the rotary wheel 230 generate the illustrated signals. FIG. 11A also shows a small portion of the peripheral ring 235 of the top of fixed base 231, as shown in FIG. 9 and described above, which surrounds conductive pads 237 and 238. Each of the four states depicted in FIG. 11A has a corresponding data point (e.g., state) in the encoder signals of FIG. 10. For instance, in FIG. 11A, the view is from under the top surface of the fixed base 231, looking upwards, with conductive pads 237 and 238, and peripheral ring 235 shown transparently, so that the alternating peripheral conductive regions 710 and non-conductive regions 720 on the bottom of the rotary wheel 230 are seen in the background. To distinguish conductive and non-conductive regions 710 and 720, respectively, conductive region 710 is shaded using diagonal lines that run from top left to bottom right (“backslash”), and non-conductive region 720 is shaded with diagonal lines that run from bottom left to top right (“frontslash”), as shown.

[0091] In the depicted example of FIG. 11A, the alternating conductive regions 710 and non-conductive regions 720 have the same shape and size. It is also noted that in the depicted example of FIG. 11A, the conductive pads 237 and 238, carrying channels A and B, respectively, on the upper surface of fixed base 231 are sized such that their pad width, W1 is one-half the width W2 of a conductive region 710 or non-conductive region 720 of the bottom of the rotary wheel, such that two of conductive pads 237 or 238 could fit within, or underneath, one conductive region 710 or non-conductive region 720. Further, the conductive pads 237 and 238 are separated from each other by two conductive pad widths W1, or one region (710, 720) width W2. The four states, as shown, indicate an anticlockwise rotation of the rotary wheel 230 relative to the fixed base 231. Accordingly, because, as noted, the view is from underneath the upper surface of the fixed base 231 looking into the bottom of the rotary wheel 230, it appears that the conductive pads 237 and 238, carrying channels A and B respectively, while in reality stationary, are moving anti-clockwise relative to the bottom of the rotary wheel 230.

[0092] Beginning with State 1 1110, the conductive pad 237 (e.g., the channel A pad) of the upper surface of the fixed base 231 is coupled to a conductive region 710A of the bottom surface of the rotary wheel 230, but the conductive pad 238 (e.g., the channel B pad) is not, being underneath a non-conductive region 720B of the bottom surface of the rotary wheel 230, as shown. Thus, in terms of the encoder signals of FIG. 10, which follow the convention that “coupled to a conductive region 710”=1, and “coupled to a non-conductive region 720”=0, channel A has a 1 and

channel B a 0, or an overall (A,B) value of (1,0). At State 2 1120, which indicates a one pad width W1 turn (which is a one half of a conductive or non-conductive region width W2 turn) of the rotary wheel 230 to the right, moving the conductive pad 237 over to the left under a next non-conductive region 720A, and moving the conductive pad 238 to be under the left side of non-conductive region 720B, now neither the A pad nor the B pad is coupled to a conductive region 710, and thus both channels A and B have a value of 0, or an overall (A,B) value of (0,0). The change from (A,B)=(1,0) to (0,0) is shown in FIG. 10 in the example anticlockwise rotation sequence 1020 as the third and fourth data points in the sequence. At State 3 1130, the conductive pad 237 has now moved by a single W1 turn to the left to be under the left side of non-conductive region 720A, and thus the A pad is still not coupled, but the B pad has moved one W1 turn to be underneath the right side of conductive region 710A, and now is coupled. Thus, channel A has a 0 value and channel B a 1 value, for an overall (0,1) value. Finally, at State 4 1140, the pads A and B have moved another single W1 turn to the left, corresponding to the rotary wheel above having turned another W1 turn to the right. Now both the conductive pad 237 and the conductive pad 238 are coupled to conductive regions of the rotary wheel underside. The conductive pad 237 has moved to the right side of conductive region 710B, and the conductive pad 238 has moved to the left side of conductive region 710A, and thus both channels A and B have values of 1, for an overall (A,B)=(1,1).

[0093] Thus, the progression of data points (A,B) through the four states of FIG. 11A is from (1,0) to (0,0) to (0,1) to (1,1). As shown at 1030 of FIG. 10, this sequence indicates an anticlockwise rotation. As noted above, it is here assumed that when a conductive pad 237 or 238 of the fixed base 231 is aligned with a conductive region 710 of the rotary wheel 230’s underside, its signal value=1, and when the conductive pad 237 or 238 is aligned with a non-conductive region 720 of the underside of rotary wheel 230, its signal value=0. In alternate examples, the inverse convention may be used.

[0094] In some examples, there is a relationship between the widths of conductive pads 237 and 238 (which have the same width, W1), and the widths of a conductive region 710 or non-conductive region 720 (which have the same width, W2). In some instances, it is the relative widths of W1 and W2 that determine the resolution with which rotations of the rotary wheel 230 relative to the fixed base 231 may be detected. In one such instance, as shown in FIG. 11A, the width W1 of each of conductive pads 237 and 238 is one half the width W2 of an underside conductive region 710 or a non-conductive region 720, respectively. Thus, in such instances, a change in rotation as a conductive pad 237 or 238 moves a W1 step may be detected. This is because in a W1 sized step a conductive pad 237 or 238 either moves from being under one side of a conductive region 710 or a non-conductive 720 to the other side of that region, as is shown in FIG. 11A for conductive pad 238 moving from one side of non-conductive region 720B to the other side of that region, between State 1 and State 2, or in a W1 sized step it moves from a second side of a conductive region 710 or a non-conductive 720, to a first side of an adjacent region of the other type, as shown, for example, in FIG. 11A for conductive pad 237 moving from the second side of conductive region 710A to the first side of non-conductive region 720A.

[0095] FIG. 11B illustrates an example distance 1160 between conductive pads 237 and 238 of fixed base 231 in terms of the W1 conductive pad width. The point of view here in FIG. 11B is now from underneath the entire fixed base 231, looking upwards into essentially FIGS. 4C and 5, where the three sensing electrodes 411, 410 and 420 of the bottom surface of the fixed base, and the two conductive pads 237 and 238 on the top surface of the fixed base, are all shown in transparent mode. As shown, there are sixteen conductive pad width W1 divisions between conductive pads 237 and 238. There are seven conductive/non-conductive regions 710, 720 between them, of width W2 each, as well as two additional W1 width regions, one to the right of the conductive pad 237 and the other to the left of the conductive pad 238. Conductive pads 237 and 238 are positioned above their corresponding coupling sensing electrodes 411 and 410, respectively, on the underside of the fixed base. The distancing of the conductive pads 237 and 238 by a distance equal to 16W1 is so as to reduce parasitic coupling from other neighboring sensing pixels. Thus, in some examples, because coupling sensing electrodes 411 and 410 have a specific location in alignment to the grid, as shown in FIG. 4C and described above, the conductive pads 237 and 238 are restricted to certain areas. In the example configuration of FIG. 11B, neither of the two conductive pads 237 and 238 are coupled to a conductive region of the underside of the rotary wheel. As shown, both are coupled to non-conductive regions 720. However, one turn to the right would move conductive pad 238, carrying the B channel, to couple to an adjacent conductive region, or, alternatively, one turn to the left would move conductive pad 237, carrying the A channel, to couple to an adjacent conductive region 710.

[0096] FIG. 12 is a flowchart of an exemplary process for determining a direction of rotation of the rotatable knob interface according to one or more examples of the present disclosure. The process 1200 may be performed by the electronic device 100 and in particular, the processing system 110 shown in FIG. 1. However, it will be recognized that an input device that includes additional and/or fewer components as shown in FIG. 1 may be used to perform process 1200, that any of the following blocks may be performed in any suitable order, and that the process 1200 may be performed in any suitable environment. The descriptions, illustrations, and processes of FIG. 12 are merely exemplary and the process 1200 may use other descriptions, illustrations, and processes for determining a direction of rotation of a rotatable knob interface 150.

[0097] As will be described below, aspects of the present disclosure provide processes 1200, 1300, and 1700 for the processing system 110 to determine an initial state of the rotatable knob interface 150 using a baseline ADC frame and one or more additional frames. For instance, the processing system 110 may determine the initial state upon start-up (e.g., the electronic device 100 may be associated with a vehicle, and the processing system 110 may determine the initial state upon and/or during start-up of the vehicle). Using the initial state of the rotatable knob interface 150 (e.g., if the initial state is (1,0) or (0,1)) and the detected first rotational state (1,1), the processing system 110 determines the direction of rotation of the rotatable knob interface 150 even if the user rotates the rotatable knob interface 150 by a single rotational state. For instance, as shown in FIG. 10, if the initial state is (1,0) and the detected

state after the user rotates the rotatable knob interface 150 is (1,1), then the processing system 110 determines the direction of rotation of the rotatable knob interface 150 is clockwise. Similarly, if the initial state is (0,1), then the processing system 110 determines the direction of rotation is anticlockwise.

[0098] In operation, at block 1202, based on providing reference signals (e.g., ground signals) to a set of alternative electrodes, the processing system 110 obtains a baseline ADC frame indicating ADC values associated with a set of rotational electrodes. As mentioned above, the rotational electrodes may be the second set of sensing electrodes 410 and 411 shown in FIGS. 4A, 4C, and 5. The rotational electrodes may be electrically connected to the conductive pads 237 and 238, which are coupled to the outer peripheral ring with the conductive regions 710 and the non-conductive regions 720. The processing system 110 may provide one or more signals (e.g., the sensing signals and/or the reference signals/ground signals) to the first, second, and/or third sets of electrodes (e.g., electrodes 410, 411, 420, and/or 430 of FIGS. 4A-4C and 5) as well as other sets of electrodes such as the alternative electrodes. The alternative electrodes may be any electrodes within the grid of electrodes that are not the electrodes 410, 411, 420, and/or 430. For instance, referring to FIG. 4A, in some examples, the alternative electrodes may be the electrodes in the column 412 and/or one or more electrodes in the first two columns (e.g., the column of electrodes to the left of the sensing electrodes 410, 411, and 420).

[0099] As such, at block 1202, the processing system 110 provides the reference signals to the alternative electrodes such as one or more of the electrodes in the first two columns of FIG. 4A. Further, the processing system 110 provides sensing signals to at least some of the other grid 401 of electrodes, and obtains resulting signals based on providing the sensing signals and the reference signals. The processing system 110 generates a raw image such as a baseline ADC frame indicating the ADC values of the resulting signals that were obtained based on providing the sensing signals and the reference signals. For instance, the baseline ADC image may indicate the resulting signals from the grid 401 of electrodes including the rotational sensing electrodes 410 and 411.

[0100] FIG. 14 shows a baseline ADC frame indicating ADC values according to one or more examples of the present disclosure. For instance, the baseline ADC frame 1400 may be obtained using the electrodes shown in FIG. 4A. In particular, the baseline ADC frame 1400 shows the grid 401 of electrodes, including the rotational sensing electrodes 410 and 411 as well as the first set of electrodes 430 (e.g., the knob ground electrodes, the reference signal electrodes, or the reference ground electrodes), except the baseline ADC frame 1400 is rotated 90 degrees counter-clockwise from the electrodes shown in FIG. 4A. For example, the circle denoted as 1402 may be the exterior circumference of the fixed base 231. The electrodes for the first knob sensing pad 1404 may be the sensing electrode 410 from FIG. 4A (e.g., the sensing electrode 410 from FIG. 4A may include two separate electrodes), the electrodes for the second knob sensing pad 1406 may be the sensing electrode 411 from FIG. 4A, and the electrodes for the knob ground (GND) pad 1414 may be the first set of electrodes 430 from FIG. 4A. In addition, the alternative electrodes (e.g., the electrodes in the first two columns of FIG. 4A) are

shown as the alternative electrodes **1408**. Thus, at block **1202**, the processing system **110** provides the reference signals to the alternative electrodes **1408** and sensing signals to the other electrodes such as the electrodes for the first knob sensing pad **1404** and the electrodes for the second knob sensing pad **1406**. Based on providing the reference signals and the sensing signals, the baseline ADC frame **1400** shows the ADC values of the resulting signals for the different electrodes such as the electrodes for the first knob sensing pad **1404** (e.g., the rotational sensing electrodes **410**) and the electrodes for the second knob sensing pad **1406** (e.g., the rotational sensing electrodes **411**). In addition, the baseline ADC frame **1400** shows the resulting signals of the knob ground pad **1414** (e.g., the first set of electrodes **430**) including two specific electrodes—a first knob fall detection electrode **1410** and a second knob fall detection electrode **1412**. The first and second knob fall detection electrodes will be described in more detail in process **1300** below.

[0101] Returning back to FIG. **12**, based on providing the sensing signals and the reference signals, the processing system **110** obtains the resulting signals and generates a baseline ADC frame. The baseline ADC frame may indicate the ADC values (e.g., raw ADC values) of the rotational electrodes (e.g., rotational sensing electrodes **410/411** and/or the knob sensing pad electrodes **1404/1406**).

[0102] At block **1204**, based on providing the reference signals to a first subset of a set of reference signal electrodes, the processing system **110** obtains a first ADC frame indicating ADC values associated with one or more first rotational electrodes from the set of rotational electrodes. For example, the processing system **110** provides the reference signals to a first subset (e.g., a portion) of a set of reference signal electrodes (e.g., the first set of electrodes **430** shown in FIG. **4A**) and sensing signals to one or more other electrodes such as first rotational electrodes (e.g., rotational sensing electrode **410** shown in FIG. **4A**) from the set of rotational electrodes (e.g., the rotational sensing electrodes **410** and **411**). The processing system **110** receives resulting signals based on providing the reference signals and the sensing signals. The processing system **110** generates a raw image such as a first ADC frame based on the resulting signals.

[0103] FIG. **15** shows a first ADC frame indicating ADC values according to one or more examples of the present disclosure. For instance, the first ADC frame **1500** may be obtained using the electrodes shown in FIG. **4A**. The exterior circumference of the fixed base **231** (e.g., circle **1402**), the knob sensing pad **1404**, the knob sensing pad **1406**, the knob ground pad **1414**, and the first knob fall detection electrode **1410** from the baseline ADC frame **1400** are also shown in the first ADC frame **1500**. Further, a portion or a first subset of electrodes **1502** from the knob ground pad **1414** (e.g., a first subset of electrodes from the first set of electrodes **430**) is also shown in the first ADC frame **1500**. The processing system **110** provides the reference signals to the first subset of electrodes **1502** from the knob ground pad **1414** and sensing signals to other electrodes such as the electrodes for the first knob sensing pad **1404** (e.g., the rotational sensing electrodes **410**). Additionally, and/or alternatively, the processing system **110** provides sensing signals to the other electrodes in the same column as the electrodes for the knob sensing pad **1404** such as the first knob fall detection electrode **1410**. Based on providing the reference

signals and the sensing signals, the processing system **110** obtains resulting signals, and generates the first ADC frame, which indicates the raw ADC values for electrodes of the knob sensing pad **1404**.

[0104] In other words, returning back to FIG. **12**, based on providing the sensing signals and the reference signals, the processing system **110** obtains the resulting signals and generates a first ADC frame. The first ADC frame may indicate the ADC values (e.g., raw ADC values) of the first rotational electrodes (e.g., rotational sensing electrode **410** and/or the knob sensing pad **1404** electrodes).

[0105] At block **1206**, based on providing the reference signals to a second subset of a set of reference signal electrodes, the processing system **110** obtains a second ADC frame indicating ADC values associated with one or more second rotational electrodes from the set of rotational electrodes. For example, the processing system **110** provides the reference signals to a second subset (e.g., a portion) of a set of reference signal electrodes (e.g., the first set of electrodes **430** shown in FIG. **4A**) and sensing signals to one or more other electrodes such as second rotational electrodes (e.g., rotational sensing electrode **411** shown in FIG. **4A**) from the set of rotational electrodes (e.g., the rotational sensing electrodes **410** and **411**). The processing system **110** receives resulting signals based on providing the reference signals and the sensing signals. The processing system **110** generates a raw image such as a second ADC frame based on the resulting signals.

[0106] FIG. **16** shows a second ADC frame indicating ADC values according to one or more examples of the present disclosure. For instance, the second ADC frame **1600** may be obtained using the electrodes shown in FIG. **4A**. The exterior circumference of the fixed base **231** (e.g., circle **1402**), the knob sensing pad **1404**, the knob sensing pad **1406**, the knob ground pad **1414**, and the second knob fall detection electrode **1412** from the baseline ADC frame **1400** are also shown in the second ADC frame **1600**. Further, a portion or a second subset of electrodes **1602** from the knob ground pad **1414** (e.g., a second subset of electrodes from the first set of electrodes **430**) is also shown in the second ADC frame **1600**. The second subset of electrodes **1602** may include one or more electrodes from the first subset of electrodes **1502** shown in FIG. **15**, but may further include one or more electrodes that were not included in the first subset of electrodes **1502**. The processing system **110** provides the reference signals to the second subset of electrodes **1602** from the knob ground pad **1414** and sensing signals to other electrodes such as the electrodes for the second knob sensing pad **1406** (e.g., the rotational sensing electrodes **411**). Additionally, and/or alternatively, the processing system **110** provides sensing signals to the other electrodes in the same column as the electrodes for the knob sensing pad **1406** such as the second knob fall detection electrode **1412**. Based on providing the reference signals and the sensing signals, the processing system **110** obtains resulting signals, and generates the second ADC frame, which indicates the raw ADC values for electrodes of the knob sensing pad **1406**.

[0107] Returning back to FIG. **12**, based on providing the sensing signals and the reference signals, the processing system **110** obtains the resulting signals and generates a second ADC frame. The second ADC frame may indicate the ADC values (e.g., raw ADC values) of the second

rotational electrodes (e.g., rotational sensing electrode **411** and/or the knob sensing pad **1406** electrodes).

[0108] At block **1208**, the processing system **110** determines an initial state of the rotatable knob interface using the baseline ADC frame, the first ADC frame, and the second ADC frame. For example, using the first ADC frame (e.g., first ADC frame **1500**), the processing system **110** determines the ADC values associated with the first rotational electrodes (e.g., the rotational sensing electrodes **410** and/or the first knob sensing pad **1404**). Further, using the baseline ADC frame (e.g., baseline ADC frame **1400**), the processing system **110** further determines the ADC values associated with the first rotational electrodes. The processing system **110** determines a difference between the ADC values for the first rotational electrodes from the first ADC frame (e.g., first ADC values for the first rotational electrodes) and the ADC values for the first rotational electrodes from the baseline ADC frame (e.g., second ADC values for the first rotational electrodes). The difference (e.g., a first knob sensing pad value indicating an ADC value change for the first rotational electrodes) may be compared with a threshold to determine the initial state of the rotatable knob interface. For example, based on comparing the first knob sensing pad value for the first rotational electrodes with a threshold value, the processing system **110** determines whether the state of the first rotational electrodes (e.g., sensing electrodes **410**) is in a “0” state or a “1” state.

[0109] Further, using the second ADC frame (e.g., second ADC frame **1600**) and the baseline ADC frame, the processing system **110** determines a difference (e.g., a second knob sensing pad value indicating an ADC value change for the second rotational electrodes) between the ADC values for the second rotational electrodes (e.g., the rotational sensing electrodes **411** and/or the second knob sensing pad **1406**) from the second ADC frame and the ADC values for the second rotational electrodes from the baseline ADC frame. Based on comparing the second knob sensing pad value for the second rotational electrodes with a threshold value, the processing system **110** determines whether the state of the second rotational electrodes (e.g., sensing electrodes **411**) is in a “0” state or a “1” state.

[0110] In other words, based on determining differences, changes, or deltas between the ADC values from the first/second frame and the ADC values from the baseline frame for the first/second rotational electrodes, the processing system **110** may determine an initial state of the rotatable knob interface such as the states (0,0), (1,0), (1,1), or (0,1) that are described above. For example, based on the difference being below a threshold (e.g., the ADC values from the first/second frame and the baseline frame are closer together), the processing system **110** may determine the rotational electrode (e.g., sensing electrode **410** or **411**) is in a “0” state. Based on the difference being above a threshold (e.g., the ADC values from the first/second frame and the baseline frame are farther apart), the processing system **110** may determine the rotational electrode (e.g., sensing electrode **410** or **411**) is in a “1” state.

[0111] For instance, because the first rotational sensing electrode **410** (e.g., the electrodes for the knob sensing pad **1404**) may have a conductive path with the first set of electrodes **430** (e.g., knob ground pad **1414**) inside the rotatable knob interface **150** when the first rotational electrode sensing **410** is in a “1” state, the first ADC frame described above (e.g., the first ADC frame **1500**), when

applying the sensing signal to the first rotational sensing electrode **410** while applying the reference signal to the first set of electrodes **430**, may cause the ADC conversion result at the first rotational sensing electrode **410** to become different. On the other hand, based on the first rotational electrode **410** being in a “1” state, the ADC baseline frame described above (e.g., the baseline ADC frame **1400**), when applying the sensing signals to both first rotational sensing electrode **410** and the first set of electrodes **430**, and applying the reference signal to the alternative electrodes **1408**, the ADC conversion results at the first rotational sensing electrode **410** might not become different.

[0112] In contrast, when the first rotational sensing electrode **410** is in a “0” state, the conductive path may not exist between the first rotational sensing electrode **410** and the first set of electrodes **430** within the rotatable knob interface **150**. The first ADC frame **1500**, when applying the sensing signal to the first rotational sensing electrode **410** while applying the reference signal to the first set of electrodes **430**, may not cause the ADC conversion result to be different at the first rotational sensing electrode **410**. Further, based on the first rotational sensing electrode **410** being in a “0” state, the ADC baseline frame **1400**, when applying the sensing signals to both first rotational sensing electrode **410** and the first set of electrodes **430**, and applying the reference signal to the alternative electrodes **1408**, the ADC conversion results at the first rotational sensing electrode **410** might also not be different.

[0113] At block **1210**, based on the rotatable knob interface being rotated from the initial state to a first rotational state, the processing system **110** obtains, using the set of rotational electrodes (e.g., sensing electrodes **410/411** and/or the electrodes for the knob sensing pad **1404/1406**), first rotational state resulting signals indicating first rotational state of the rotatable knob interface **150**. The first rotational state of the rotatable knob interface may be the first state of the conductive pads **237**, **238** after the user rotates the rotatable knob interface **150**. For example, after the user rotates the rotatable knob interface **150** by one state (e.g., a single resolution, detent, and so on), the processing system **110** obtains first rotational state indicating the states (e.g., the first rotational state) of the conductive pads **237**, **238**. For instance, the user may rotate the rotatable knob interface **150** to change a setting (e.g., volume, temperature, and/or fan). The processing system **110** may provide one or more signals to the electrodes that interact with the rotatable knob interface **150**. For instance, as described above, the processing system **110** may provide reference signals to the first set of electrodes (e.g., reference signal electrodes such as the first set of electrodes **430**). The reference signals may be a ground signal signals (e.g., a configurable DC output provided by the processing system **110**). Further, the processing system **110** may provide sensing signals to the first and/or the second set of electrodes such as the rotational sensing electrodes **410**, **411**. Depending on the connection between the conductive pads **237**, **238** and the regions of the outer peripheral ring **701** (e.g., the conductive and non-conductive regions **710**, **720**), the processing system **110** may obtain different resulting signals. For instance, referring back to FIG. 11A and state 1, because the conductive pad **237** (e.g., channel A) is coupled to a conductive region **710A** of the outer peripheral ring **701**, the processing system **110** obtains a state for channel A of “1”. Because the conductive pad **238** (e.g., channel B) is coupled to a non-conductive region

720B, the processing system 110 obtains a state for channel B of “0”. As such, the processing system 110 determines the state (e.g., the first rotational state) as (1,0). Similarly, for state 3, if the conductive pad 237 is coupled to the non-conductive region 720A and the conductive pad 238 is coupled to the conductive region 710A, the processing system 110 determines the state as (0,1).

[0114] In some instances, the processing system 110 may provide the reference signals (e.g., a ground signal) and the sensing signals (e.g., a sine wave from a guard amplifier (AMP)) at the same time or at substantially the same time. In some examples, the processing system 110 may provide the signals (e.g., both the reference and sensing signals) at particular time intervals. For instance, the processing system 110 may provide the signals every 8.33 milliseconds (ms), which may be half of the display frame. The processing system 110 may provide the signals regardless of whether there is user input.

[0115] At block 1212, the processing system 110 determines a rotational direction of the rotatable knob interface using the determined initial state and the first rotational state resulting signals. For instance, the processing system 110 may determine the first rotational state using the first rotational state (e.g., a state of (1,1)). The processing system 110 may further determine an initial state of the rotatable knob interface 150 as described in block 1208 (e.g., a state of (1,0)). Then, referring to FIG. 10, using these determined states, the processing system 110 determines a rotational direction of the rotatable knob interface 150. For instance, because the transition from the initial state to the first rotational state of the rotatable knob interface is from (1,0) to (1,1), the processing system 110 determines the rotational directional of the rotatable knob interface as a clockwise rotation. Likewise, if the initial state is (0,1) and because the transition from the initial state to the first rotational state of the rotatable knob interface is from (0,1) to (1,1), the processing system 110 determines the rotational directional of the rotatable knob interface as an anticlockwise rotation.

[0116] At block 1214, the processing system 110 adjusts a setting using the determined rotational direction. For instance, the electronic system and/or the electronic device 100 may include and/or control a plurality of settings. For example, the electronic system may be part of a vehicle and the electronic device 100 may control a plurality of settings for the vehicle including, but not limited to, fan, temperature, volume, and so on. Using the determined rotational direction, the processing system 110 may determine whether to increase or decrease the setting. Furthermore, in some instances, the user may seek to adjust a setting by a single state (e.g., a single resolution, detent, and so on). As such, using process 1200, the processing system 110 may determine the rotational direction even though the user has adjusted the setting by a single state by using the initial state prior to rotation of the rotatable knob interface 150. The processing system 110 may then adjust the setting using the determined rotational direction.

[0117] FIGS. 13 and 17 will describe the processing system 110 determining the initial state of the rotatable knob interface 150 (e.g., blocks 1202 and 1204) in more detail. For example, FIG. 13 is another flowchart of an exemplary process for determining a direction of rotation of the rotatable knob interface according to one or more examples of the present application. The process 1300 may be performed by the electronic device 100 and in particular, the processing

system 110 shown in FIG. 1. However, it will be recognized that any of the following blocks may be performed in any suitable order, and that the process 1300 may be performed in any suitable environment. The descriptions, illustrations, and processes of FIG. 13 are merely exemplary and the process 1300 may use other descriptions, illustrations, and processes for determining the direction of rotation of the rotatable knob interface 150.

[0118] At block 1302, the processing system 110 performs blocks 1202 through 1206 as described above in FIG. 12. For example, as mentioned above, by performing these blocks, the processing system 110 obtains the baseline ADC frame, the first ADC frame, and the second ADC frame. In addition to the baseline ADC frame, the first ADC frame, and/or the second ADC frame indicating the raw ADC values for the electrodes of the knob sensing pads 1404 and/or 1406, the baseline ADC frame, the first ADC frame, and/or the second ADC frame also show the raw ADC values for the first and second knob fall detection electrodes 1410 and 1412. For instance, the first ADC frame and the baseline ADC frame show the ADC values for the first knob fall detection electrode 1410 and the second ADC frame and the baseline ADC frame show the ADC values for the second knob fall detection electrode 1412.

[0119] At block 1304, the processing system 110 determines one or more knob fall-off values using the baseline ADC frame, the first ADC frame, and the second ADC frame. The knob fall-off values may indicate differences between the first and second knob fall detection electrodes 1410 and 1412 from the baseline frame to the first or second ADC frames. For example, the processing system 110 determines a first knob fall-off value for the first knob fall detection electrode 1410 by subtracting the ADC value of the first knob fall detection electrode 1410 from the first ADC frame 1500 from the ADC value of the first knob fall detection electrode 1410 from the baseline ADC frame 1400. Similarly, the processing system 110 determines a second knob fall-off value for the second knob fall detection electrode 1412 by subtracting the ADC value of the second knob fall detection electrode 1412 from the second ADC frame 1600 from the ADC value of the second knob fall detection electrode 1412 from the baseline ADC frame 1400.

[0120] At block 1306, the processing system 110 determines a first knob sensing pad value and a second knob sensing pad value based on using the baseline ADC frame, the first ADC frame, and the second ADC frame. The first knob sensing pad value may be a difference between the ADC values for the first rotational electrodes from the first ADC frame and the ADC values for the first rotational electrodes from the baseline ADC frame. The second knob sensing pad value may be a difference between the ADC values for the second rotational electrodes from the second ADC frame and the ADC values for the second rotational electrodes from the baseline ADC frame. The determination of the first and second knob sensing pad values are described above.

[0121] At block 1308, the processing system 110 performs a knob fall-off detection based on comparing a knob fall-off threshold with an aggregation of the one or more knob fall-off values, the first knob sensing pad value, and the second knob sensing pad value. For instance, based on the one or more knob fall-off values, the first knob sensing pad value, and the second knob sensing pad value, the process-

ing system 110 determines whether the rotatable knob interface 150 has fallen off the display (e.g., the display screen). For example, as mentioned above, the electronic device 100 with the rotatable knob interface 150 may be affixed to a display screen. In some instances, due to use, the rotatable knob interface 150 might not be completely affixed to the display screen anymore and process 1300 may be used to determine whether the rotatable knob interface 150 has detached from the display screen.

[0122] For example, the processing system 110 may aggregate the knob fall-off values (e.g., the first and second knob fall-off values associated with the first/second fall detection electrodes 1410 and 1412), the first knob sensing pad value associated with the first rotational sensing electrode 410, and/or the second knob sensing pad value associated with the second rotational electrode 411. The processing system 110 may compare the aggregated values with one or more thresholds. Based on the aggregated values exceeding the threshold (e.g., greater than the threshold), the processing system 110 may proceed to the next block (e.g., block 1310). Based on the aggregated values not exceeding the threshold (e.g., less than the threshold), the processing system 110 might not proceed to the next block. Instead, the processing system 110 may provide an alert (e.g., failure alert) on a display device indicating that the rotatable knob interface 150 may have detached from the display screen.

[0123] At block 1310, based on performing the knob fall-off detection (e.g., the aggregated values exceeding the threshold), the processing system 110 performs blocks 1208-1214. For example, using the first and second knob sensing pad values, the processing system 110 determines the initial state of the rotatable knob interface 150, obtains the first rotational state resulting signals, determines the rotational direction of the rotatable knob interface, and/or adjusts a setting using the determined rotational direction.

[0124] For example, FIG. 17 is yet another flowchart of an exemplary process for determining a direction of rotation of the rotatable knob interface according to one or more examples of the present application. The process 1700 may be performed by the electronic device 100 and in particular, the processing system 110 shown in FIG. 1. However, it will be recognized that any of the following blocks may be performed in any suitable order, and that the process 1700 may be performed in any suitable environment. The descriptions, illustrations, and processes of FIG. 17 are merely exemplary and the process 1700 may use other descriptions, illustrations, and processes for determining the direction of rotation of the rotatable knob interface 150.

[0125] At block 1702, the processing system 110 performs blocks 1202 through 1208 as described above in FIG. 12 and/or blocks 1304-1308 as described in FIG. 13. For example, as mentioned above, by performing blocks 1202-1208, the processing system 110 obtains the baseline ADC frame, the first ADC frame, the second ADC frame, and determines an initial state of the rotatable knob interface 150. In some examples, blocks 1304-1308 are performed as well. When performed, the processing system 110 performs the knob fall-off detection.

[0126] At block 1704, based on providing reference signals to the set of reference signal electrodes and sensing signals to the set of rotational electrodes, the processing system 110 obtains a touch frame indicating ADC values associated with the set of rotational electrodes. For example, in blocks 1202-1206, the processing system 110 provides the

reference signals to a set of alternative electrodes (e.g., the alternative electrodes 1408) to obtain the baseline ADC frame, the reference signals to a first and second subset of the reference signal electrodes (e.g., the first subset of electrodes 1502 and the second subset of electrodes 1602) to obtain the first and second ADC frames. In block 1704, the processing system 110 provides the reference signals to all of the reference signal electrodes (e.g., the first set of electrodes 430 and/or the electrodes of the knob ground pad 1414). The processing system 110 further provides the sensing signals to the electrodes of the knob sensing pads 1404 and 1406 (e.g., the rotational sensing electrodes 410, 411). Based on providing the reference signals and the sensing signals, the processing system 110 obtains a touch frame indicating ADC values associated with the set of rotational electrodes (e.g., the rotational sensing electrodes 410, 411 and/or the electrodes of the knob sensing pads 1404 and 1406).

[0127] At block 1706, the processing system 110 determines a first knob baseline value for the first knob sensing pad based on the touch frame and the determined initial state for the first knob sensing pad. For example, based on the determined initial state for the first knob sensing pad (e.g., a state of "0" or a state of "1") and the ADC value for the first rotational electrode (e.g., the rotational sensing electrode 410 and/or the electrodes of the knob sensing pad 1404), the processing system 110 determine the first knob baseline value. For instance, based on the determined state being "0", the processing system 110 may determine the first knob baseline value as the ADC value for the first rotational electrode from the touch frame. Based on the determined state being "1", the processing system 110 may determine the first knob baseline value as the ADC value for the first rotational electrode from the touch frame minus a pre-defined value (e.g., "200"). As such, if the determined state is "1", the processing system 110 may subtract the ADC value for the first rotational electrode from the touch frame (e.g., "500") minus a pre-defined value (e.g., "200") to determine the first knob baseline value (e.g., "300").

[0128] At block 1708, the processing system 110 determines a second knob baseline value for the second knob sensing pad based on the touch frame and the determined initial state for the second knob sensing pad. For example, based on the determined initial state for the second knob sensing pad (e.g., a state of "0" or a state of "1") and the ADC value for the second rotational electrode (e.g., the rotational sensing electrode 411 and/or the electrodes of the knob sensing pad 1406), the processing system 110 determine the second knob baseline value. For instance, based on the determined state being "0", the processing system 110 may determine the second knob baseline value as the ADC value for the second rotational electrode from the touch frame. Based on the determined state being "1", the processing system 110 may determine the second knob baseline value as the ADC value for the second rotational electrode from the touch frame minus a pre-defined value (e.g., "200"). As such, if the determined state is "1", the processing system 110 may subtract the ADC value for the second rotational electrode from the touch frame (e.g., "500") minus a pre-defined value (e.g., "200") to determine the second knob baseline value (e.g., "300").

[0129] At block 1710, the processing system 110 performs blocks 1210-1214 based on the first knob baseline value and the second knob baseline value. For example, if the raw

ADC value for the first rotational electrode is substantially different from the first knob baseline value (e.g., the difference between the two is greater than a threshold), the processing system 110 may determine the state of the first rotational electrode is “1”. If it is not substantially different from the first knob baseline value (e.g., the difference between the two is less than a threshold), the processing system 110 may determine the state of the first rotational electrode is “0”. Similarly, if the raw ADC value for the second rotational electrode is substantially different from the second knob baseline value (e.g., the difference between the two is greater than a threshold), the processing system 110 may determine the state of the second rotational electrode is “1”. If it is not substantially different from the second knob baseline value (e.g., the difference between the two is less than a threshold), the processing system 110 may determine the state of the second rotational electrode is “0”.

[0130] In other words, after determining the initial states, process 1700 may be used to determine the knob baseline data (e.g., data indicating the first and second knob baseline values) even if the rotatable knob interface is at an “ON” state (e.g., state of “1”) based on subtracting a pre-defined value, which may be knob interface 150 dependent, from the ADC value from the touch frame. The processing system 110 may use the baseline data to begin determining the rotatable knob interface 150 rotations after a single detent change.

[0131] FIGS. 18A and 18B show knob initial state detection graphical representations for different knob states according to one or more examples of the present disclosure. For example, FIGS. 18A and 18B show the graphical representations 1800, 1810, 1820, and 1830, which are for different knob states. As shown, a threshold 1840 is used to determine whether the knob state is “1” or “0” (e.g., if above the threshold, it is “1”). The ADC value changes were detected at different temperature settings.

[0132] FIGS. 19A and 19B show knob fall-off diagnosis graphical representations for different knob states according to one or more examples of the present disclosure. For example, FIGS. 19A and 19B show the graphical representations 1900, 1910, 1920, and 1930, which are for different knob states. As shown, a threshold 1940 is used to determine whether the knob has been detached from the display screen. Further each of the graphical representations shows two samples with knobs on the display and one sample with a knob off the display. In addition, for each of the samples, the ADC values were detected at different temperature settings.

[0133] All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0134] The use of the terms “a” and “an” and “the” and “at least one” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The use of the term “at least one” followed by a list of one or more items (for example, “at least one of A and B”) is to be construed to mean one item selected from the listed items (A or B) or any combination of two or more of the listed items (A and B), unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and

“containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention. [0135] Exemplary embodiments are described herein. Variations of those exemplary embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein.

[0136] Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

1. A method for determining a rotational direction of a rotatable knob interface, comprising:

based on providing reference signals to a set of alternative electrodes, obtaining, by a processing system, a baseline analog-to-digital (ADC) frame indicating ADC values associated with a set of rotational electrodes;

based on providing the reference signals to a first subset of a set of reference signal electrodes, obtaining, by the processing system, a first ADC frame indicating ADC values associated with one or more first rotational electrodes from the set of rotational electrodes;

based on providing the reference signals to a second subset of the set of reference signal electrodes, obtaining, by the processing system, a second ADC frame indicating ADC values associated with one or more second rotational electrodes from the set of rotational electrodes;

determining, by the processing system, an initial state of the rotatable knob interface using the baseline ADC frame, the first ADC frame, and the second ADC frame;

based on the rotatable knob interface being rotated from the initial state to a first rotational state, obtaining, by the processing system and using the set of rotational electrodes, first rotational state resulting signals indicating the first rotational state of the rotatable knob interface;

determining, by the processing system, the rotational direction of the rotatable knob interface using the determined initial state of the rotatable knob interface and the first rotational state resulting signals; and

adjusting, by the processing system, a setting using the determined rotational direction.

2. The method of claim 1, wherein determining the initial state of the rotatable knob interface comprises:

- determining a first knob sensing pad value for the one or more first rotational electrodes based on the ADC values from the first frame and the ADC values from the baseline ADC frame;
 - determining a second knob sensing pad value for the one or more second rotational electrodes based on the ADC values from the second frame and the ADC values from the baseline ADC frame; and
 - determining the initial state of the rotatable knob interface based on the first knob sensing pad value and the second knob sensing pad value.
3. The method of claim 2, wherein determining the first knob sensing pad value comprises subtracting the ADC values for the one or more first rotational electrodes from the first frame from the ADC values for the one or more first rotational electrodes from the baseline ADC frame, and wherein determining the second knob sensing pad value comprises subtracting the ADC values for the one or more second rotational electrodes from the second frame from the ADC values for the one or more second rotational electrodes from the baseline ADC frame.
4. The method of claim 2, wherein determining the initial state of the rotatable knob interface is based on:
- comparing the first knob sensing pad value to a threshold; and
 - comparing the second knob sensing pad value to the threshold.
5. The method of claim 1, wherein the baseline ADC frame indicates a first ADC value for a first knob fall detection electrode and a second ADC value for a second knob fall detection electrode, wherein the first ADC frame indicates a third ADC value for the first knob fall detection electrode, wherein the second ADC frame indicates a fourth ADC value for the second knob fall detection electrode, wherein the method further comprises:
- performing a knob fall-off detection based on the first ADC value, the second ADC value, the third ADC value, and the fourth ADC value, and wherein determining the initial state of the rotatable knob interface is based on performing the knob fall-off detection.
6. The method of claim 5, wherein the method further comprises:
- determining, based on the first ADC value, the second ADC value, the third ADC value, and the fourth ADC value, a first knob fall-off value and a second knob fall-off value; and
 - determining a first knob sensing pad value and a second knob sensing pad value based on the baseline ADC frame, the first ADC frame, and the second ADC frame, wherein performing the knob fall-off detection is based on the first knob fall-off value, the second knob fall-off value, the first knob sensing pad value, and the second knob sensing pad value.
7. The method of claim 6, wherein performing the knob fall-off detection comprises:
- aggregating the first knob fall-off value, the second knob fall-off value, the first knob sensing pad value, and the second knob sensing pad value to obtain an aggregated value; and
 - determining whether the rotatable knob interface is detached from a display screen based on comparing the aggregated value with a knob fall-off threshold.
8. The method of claim 1, further comprising:

- based on providing the reference signals to the set of reference signal electrodes and sensing signals to the set of rotational electrodes, obtaining, by the processing system, a touch frame indicating ADC values associated with the set of rotational electrodes; and
- determining baseline data based on the touch frame, and wherein determining the rotational direction of the rotatable knob interface is based on using the baseline data and the first rotational state resulting signals.

9. The method of claim 8, wherein the determined initial state indicates an initial state for the one or more first rotational electrodes and an initial state for the one or more second rotational electrodes, and wherein determining the baseline data based on the touch frame comprises:

- determining a first knob baseline value for the one or more first rotational electrodes based on the touch frame and the initial state for the one or more first rotational electrodes; and
- determining a second knob baseline value for the one or more second rotational electrodes based on the touch frame and the initial state for the one or more second rotational electrodes, and wherein determining the rotational direction of the rotatable knob interface is based on using the first rotational state resulting signals, the first knob baseline value, and the second knob baseline value.

10. The method of claim 9, wherein determining the first knob baseline value comprises:

- based on the initial state for the one or more first rotational electrodes indicating “0”, determining the first knob baseline value as the ADC value for the one or more first rotational electrodes indicated by the touch frame; and
- based on the initial state for the one or more first rotational electrodes indicating “1”, determining the first knob baseline value based on subtracting the ADC value for the one or more first rotational electrodes indicated by the touch frame by a pre-defined value.

11. The method of claim 9, wherein determining the second knob baseline value comprises:

- based on the initial state for the one or more second rotational electrodes indicating “0”, determining the second knob baseline value as the ADC value for the one or more second rotational electrodes indicated by the touch frame; and
- based on the initial state for the one or more second rotational electrodes indicating “1”, determining the second knob baseline value based on subtracting the ADC value for the one or more second rotational electrodes indicated by the touch frame by a pre-defined value.

12. The method of claim 1, wherein the baseline ADC frame, the first ADC frame, and the second ADC frame are obtained when the rotatable knob interface is stationary.

13. The method of claim 1, wherein the processing system is associated with an electric system, wherein the electric system is a control system of a vehicle, and wherein determining the initial state of the rotatable knob interface comprises determining the initial state of the rotatable knob interface during start-up operations for the vehicle.

14. The method of claim 1, wherein the set of rotational electrodes comprise the one or more first rotation electrodes that are in electrical contact with a first conductive pad and the one or more second rotation electrodes that are in

electrical contact with a second conductive pad, wherein the first conductive pad and the second conductive pad are in contact with a peripheral ring comprising alternating conductive regions and non-conductive regions, and wherein the initial state and the first rotational state indicate whether the first conductive pad and the second conductive pad are in contact with the conductive region or the non-conductive region of the peripheral ring.

15. A processing system for determining a rotational direction of a rotatable knob interface, comprising:

one or more processors; and

a non-transitory computer-readable medium having processor-executable instructions stored thereon, the processor-executable instructions, when executed by the one or more processors, facilitating performance of the following:

based on providing reference signals to a set of alternative electrodes, obtaining a baseline analog-to-digital (ADC) frame indicating ADC values associated with a set of rotational electrodes;

based on providing the reference signals to a first subset of a set of reference signal electrodes, obtaining a first ADC frame indicating ADC values associated with one or more first rotational electrodes from the set of rotational electrodes;

based on providing the reference signals to a second subset of the set of reference signal electrodes, obtaining a second ADC frame indicating ADC values associated with one or more second rotational electrodes from the set of rotational electrodes;

determining an initial state of the rotatable knob interface using the baseline ADC frame, the first ADC frame, and the second ADC frame;

based on the rotatable knob interface being rotated from the initial state to a first rotational state, obtaining, using the set of rotational electrodes, first rotational state resulting signals indicating the first rotational state of the rotatable knob interface;

determining the rotational direction of the rotatable knob interface using the determined initial state of the rotatable knob interface and the first rotational state resulting signals; and

adjusting a setting using the determined rotational direction.

16. The processing system of claim **15**, wherein determining the initial state of the rotatable knob interface comprises:

determining a first knob sensing pad value for the one or more first rotational electrodes based on the ADC values from the first frame and the ADC values from the baseline ADC frame;

determining a second knob sensing pad value for the one or more second rotational electrodes based on the ADC values from the second frame and the ADC values from the baseline ADC frame; and

determining the initial state of the rotatable knob interface based on the first knob sensing pad value and the second knob sensing pad value.

17. The processing system of claim **16**, wherein determining the first knob sensing pad value comprises subtracting the ADC values for the one or more first rotational electrodes from the first frame from the ADC values for the one or more first rotational electrodes from the baseline ADC frame, and

wherein determining the second knob sensing pad value comprises subtracting the ADC values for the one or more second rotational electrodes from the second frame from the ADC values for the one or more second rotational electrodes from the baseline ADC frame.

18. The processing system of claim **15**, wherein the baseline ADC frame indicates a first ADC value for a first knob fall detection electrode and a second ADC value for a second knob fall detection electrode, wherein the first ADC frame indicates a third ADC value for the first knob fall detection electrode, wherein the second ADC frame indicates a fourth ADC value for the second knob fall detection electrode, wherein the processor-executable instructions, when executed by the one or more processors, further facilitating performance of the following:

performing a knob fall-off detection based on the first ADC value, the second ADC value, the third ADC value, and the fourth ADC value, and wherein determining the initial state of the rotatable knob interface is based on performing the knob fall-off detection.

19. The processing system of claim **15**, wherein the processor-executable instructions, when executed by the one or more processors, further facilitating performance of the following:

based on providing the reference signals to the set of reference signal electrodes and sensing signals to the set of rotational electrodes, obtaining, by the processing system, a touch frame indicating ADC values associated with the set of rotational electrodes; and

determining baseline data based on the touch frame, and wherein determining the rotational direction of the rotatable knob interface is based on using the baseline data and the first rotational state resulting signals.

20. A non-transitory computer-readable medium having processor-executable instructions stored thereon, the processor-executable instructions, when executed, facilitating performance of the following:

based on providing reference signals to a set of alternative electrodes, obtaining a baseline analog-to-digital (ADC) frame indicating ADC values associated with a set of rotational electrodes;

based on providing the reference signals to a first subset of a set of reference signal electrodes, obtaining a first ADC frame indicating ADC values associated with one or more first rotational electrodes from the set of rotational electrodes;

based on providing the reference signals to a second subset of the set of reference signal electrodes, obtaining a second ADC frame indicating ADC values associated with one or more second rotational electrodes from the set of rotational electrodes;

determining an initial state of a rotatable knob interface using the baseline ADC frame, the first ADC frame, and the second ADC frame;

based on the rotatable knob interface being rotated from the initial state to a first rotational state, obtaining, using the set of rotational electrodes, first rotational state resulting signals indicating the first rotational state of the rotatable knob interface;

determining the rotational direction of the rotatable knob interface using the determined initial state of the rotatable knob interface.

able knob interface and the first rotational state resulting signals; and
adjusting a setting using the determined rotational direction.

* * * * *